

HELM-BERT: Topology-Aware Representations for Chemically Modified Peptides

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Cite This: *J. Chem. Inf. Model.* 2026, 66, 7900–7916



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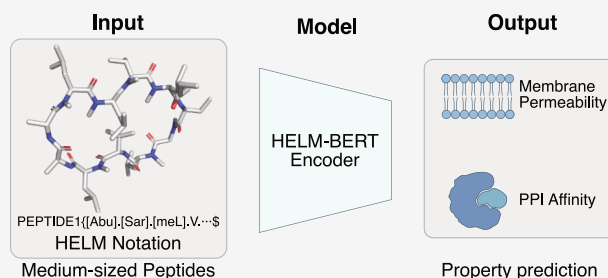


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ABSTRACT: Chemically modified and macrocyclic peptides are increasingly important therapeutics, yet current molecular representation models do not natively represent chemical modification and covalent topology in a unified way. Atom-level strings obscure macrocyclic connectivity, whereas protein sequence models cannot encode noncanonical residues and explicit cross-links. Here we pretrain an encoder-only transformer directly on Hierarchical Editing Language for Macromolecules (HELM) notation, which specifies monomer identity and connectivity. In this work, we show that the resulting representations achieve best mean performance in cyclic peptide membrane permeability prediction (random split $R^2 = 0.668$; retaining best mean performance under a Murcko scaffold split), exceeding external pretrained SMILES-based encoders. An architecture-matched SMILES control narrowed the HELM–SMILES gap under full finetuning, whereas HELM-BERT retained clearer advantages in frozen-representation settings. HELM-BERT also preserves HELM-specified macrocyclic topology in a linearly accessible form and supports competitive peptide–protein interaction prediction across complementary Propedia and ChEMBL benchmarks. More broadly, these findings suggest that pretraining directly on notation that makes structural constraints explicit offers a transferable strategy for biomolecular modalities that fall between small-molecule chemistry and protein sequence.



INTRODUCTION

Therapeutic peptides have emerged as a major drug modality bridging small molecules and biologics, with increasing clinical adoption and diversity of clinical indications.^{1,2} A central driver of this expansion is the ability to tune peptide properties through chemical modification and macrocyclization, which can improve potency, selectivity, proteolytic stability, and membrane permeability.^{3,4} However, these benefits come at a representational challenge: modified and cyclic peptides cannot be described solely as canonical amino-acid sequences, and their function depends not only on monomer composition but also on topological connectivity, including ring closures and cross-links.

Recent large-scale molecular representation models have transformed representation learning for small molecules and proteins, enabling strong transfer across tasks. For small molecules, SMILES-based language models (e.g., MoLFormer, ChemBERTa, SMILES-BERT)^{5–7} have demonstrated broad utility in property prediction. For proteins, large-scale language models trained on millions of sequences can produce representations that transfer to structure prediction and functional tasks.⁸ Yet chemically modified peptides occupy a regime that is inadequately addressed, falling between residue-only protein models and atom-level small molecule models.

SMILES⁹ represents molecules as linearized atom-and-bond sequences; for macrocycles, the ring structure is encoded implicitly through numeric ring closure markers and is not

separated from the atom stream. This makes topology a latent property that must be inferred from the string, and it complicates learning in settings where topology is a major determinant of function.^{10,11} Meanwhile, protein language models assume a residue alphabet and standard peptide bonds; noncanonical residues and many chemical modifications must be approximated, omitted, or mapped to ambiguous tokens. As a result, chemically modified peptides often require task-specific feature engineering or expensive structural pipelines (e.g., conformer sampling) rather than being handled as first-class inputs in a unified representation.

This representational gap has practical consequences for peptide drug discovery. When topology and chemical modifications are not explicit in the input, models can learn spurious correlations tied to data set-specific string conventions. Consequently, transfer across peptide tasks becomes brittle, especially when the downstream property is strongly affected by macrocyclization, cross-link patterning, or modification chemistry. A representation that makes connectivity

Received: February 11, 2026

Revised: April 28, 2026

Accepted: May 7, 2026

Published: July 15, 2026



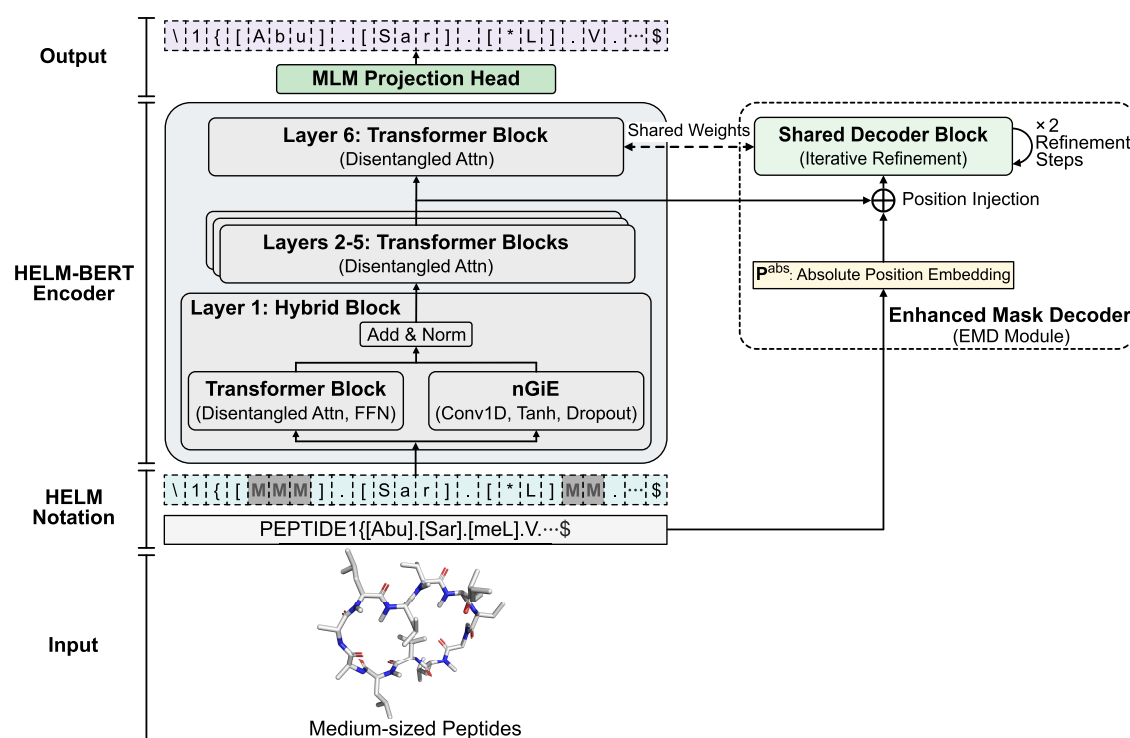


Figure 1. Overview of HELM-BERT architecture. Input peptides are converted to HELM notation, tokenized with a HELM-GPT-style compressed-character tokenizer with linearized connectivity descriptors, and subjected to span masking, where contiguous spans of tokens are masked (gray) during pretraining. The HELM-BERT encoder comprises a hybrid first layer combining disentangled self-attention with nGiE, followed by five transformer blocks with disentangled attention. The EMD receives the output of Layer 5, injects absolute position embeddings (P_{abs}), and applies two weight-tied iterative refinement steps using the same parameters as Layer 6. The MLM projection head predicts the masked tokens (green).

explicit is therefore more than a mere formatting alternative; it changes which invariances and task-relevant structural factors are accessible to learning.

HELM¹² separates the sequence of monomers from a structured connection list describing cyclization and cross-links, offering a notation that directly exposes topology. HELM has been used primarily as a descriptive standard and has recently supported generative modeling;¹³ however, a transferable encoder-only representation pretrained directly on HELM notation for downstream predictive tasks has not been established.

In this work, we introduce HELM-BERT, an encoder-only transformer pretrained with masked language modeling directly on HELM notation to learn topology-aware representations for chemically modified peptides. To capture both local chemical patterns (e.g., modification motifs) and global topological constraints (e.g., ring closures and cross-links), we incorporate key architectural components from DeBERTa,¹⁴ including disentangled attention, an Enhanced Mask Decoder (EMD), and n-gram induced encoding (nGiE). We pretrain HELM-BERT on a curated corpus of 39,079 unique peptide sequences compiled from ChEMBL, CycPeptMPDB, and Propedia after source-specific HELM processing, and evaluate transfer on two downstream tasks chosen to probe topology dependence and generalization: cyclic peptide membrane permeability prediction and peptide–protein interaction prediction. Beyond predictive performance, we quantify whether topology information explicitly specified in HELM remains linearly decodable in the embedding space. Together, these analyses test a calibrated hypothesis: pretraining on an explicit topology-aware notation yields

representations in which HELM-specified connectivity remains accessible, supporting transfer on peptide tasks that depend on topology and chemical modification.

METHODS

Model Architecture

HELM-BERT is built on the DeBERTa^{14,15} architecture (Figure 1). The backbone is a six-layer transformer encoder with hidden dimension $H = 768$ and $A = 12$ attention heads. We describe the tokenization scheme and three key components: n-gram induced encoding (nGiE), disentangled self-attention, and the Enhanced Mask Decoder (EMD).

Tokenization and Input Representation. We constructed a HELM-GPT-style compressed-character tokenizer with motif compression following HELM-GPT.¹³ Starting from the HELM-GPT character-level scheme, in which common multicharacter HELM motifs are replaced by single-character markers prior to tokenization, we made three encoder-specific extensions so that the same dictionary can be used for BERT-style MLM pretraining on HELM notation.

First, we added a dedicated [MASK] token together with [BOS] and [SEP] markers at the sequence start and end for BERT-style MLM pretraining. Second, we extended the HELM-GPT compression dictionary from two motifs ("PEPTIDE" → "/", "me" → "*") to four (adding "am" → "\t" and "ac" → "&"), covering the most frequent amide/acetyl modifiers observed in our ChEMBL/CycPeptMPDB/Propedia corpus so that these motifs remain single tokens rather than expanding into several characters. Third, we annotated every entry in the dictionary with a token category (special, canonical amino acid, HELM structural symbol, digit, other uppercase letter, lowercase letter, polymer-compressed marker, or miscellaneous) and restricted MLM span masking to nonspecial tokens via this category annotation.

The resulting vocabulary contains 78 tokens distributed across eight categories (special tokens, canonical amino acids, HELM structural symbols, digits, additional uppercase letters, lowercase letters, polymer-compressed markers, and miscellaneous); the full token-to-ID mapping is given in Table S1.

Each peptide is tokenized at the compressed-character level rather than by a true learned monomer vocabulary: canonical residue symbols are single-character tokens, whereas bracketed nonstandard monomer names are represented by their constituent characters after motif compression. The HELM connection list—which specifies pairs of connected monomers and the R-group atoms involved (e.g., PEPTIDE1, PEPTIDE1, 5:R3–10:R3 for a side-chain–side-chain cross-link between monomer 5 and monomer 10 of polymer chain 1 via the R₃ side-chain handle)—is appended to the simple-polymer section and passed through the same tokenizer. This is a linearized HELM connection representation rather than a graph-structured encoder. Because brackets, digits, and connector symbols are first-class vocabulary entries, ring closures and cross-links are preserved as explicit tokens, enabling the encoder to condition predictions on both monomer identity and connectivity patterns rather than treating connection information as out-of-vocabulary text. Rare or previously unseen monomer names can be tokenized as long as they use in-vocabulary characters; truly unseen characters are mapped to [UNK].

n-Gram Induced Encoding (nGiE). To capture local patterns in HELM token sequences, we incorporate nGiE, the first-layer 1D-convolutional local-context module of the DeBERTa-v2/v3 architecture,¹⁵ in the first encoder layer. Let $\mathbf{H}_0$ be the input embedding matrix. We compute a convolutional feature map with kernel size $k = 3$ and apply dropout and tanh nonlinearity

$$\mathbf{H}_1^{\text{conv}} = \tanh(\text{Dropout}(\text{Conv1D}(\mathbf{H}_0, k = 3))) \quad (1)$$

In parallel, we compute disentangled self-attention

$$\mathbf{H}_1^{\text{attn}} = \text{DSA}(\mathbf{H}_0) \quad (2)$$

The first-layer output is the layer-normalized residual combination

$$\mathbf{H}_1 = \text{LayerNorm}(\mathbf{H}_1^{\text{attn}} + \mathbf{H}_1^{\text{conv}}) \quad (3)$$

Disentangled Self-Attention. In disentangled self-attention, attention scores are computed as a sum of content–content, content–position, and position–content interactions

$$A_{i,j} = (\mathbf{Q}_i^c (\mathbf{K}_j^c)^T) + (\mathbf{Q}_i^c (\mathbf{K}_{\delta(i,j)}^r)^T) + (\mathbf{K}_j^c (\mathbf{Q}_{\delta(j,i)}^r)^T) \quad (4)$$

Where $\mathbf{Q}^c$ and $\mathbf{K}^c$ are content query and key vectors, $\mathbf{Q}^r$ and $\mathbf{K}^r$ are relative-position query and key vectors, and $\delta(i,j)$ denotes the clipped relative distance between tokens i and j . This formulation enables explicit modeling of relative-position effects that are critical when interpreting HELM sequence and connectivity tokens jointly.

Enhanced Mask Decoder (EMD). Following DeBERTa,¹⁴ we use an Enhanced Mask Decoder (EMD) to improve masked-token reconstruction. Let $\mathbf{H}_{L-1}$ be the output of the penultimate transformer layer. We initialize the decoder query with absolute positional embeddings

$$\mathbf{Q}^{(0)} = \mathbf{H}_{L-1} + \mathbf{P}^{\text{abs}}, \mathbf{K} = \mathbf{V} = \mathbf{H}_{L-1} \quad (5)$$

We then apply two iterative refinement steps using a weight-tied transformer block (sharing parameters with the final encoder block)

$$\mathbf{Q}^{(t)} = \text{TransformerBlock}_L(\mathbf{Q}^{(t-1)}, \mathbf{K}, \mathbf{V}), \quad t = 1, 2 \quad (6)$$

The final refined representation is projected to vocabulary logits for MLM prediction.

Pretraining Data Sources and Processing

We collected peptide data from three sources: ChEMBL v35¹⁶ (22,045 entries with HELM notation), CycPeptMPDB v1.2¹⁷ (8,466 cyclic peptides with experimental membrane permeability and HELM notation), and Propedia v2.3¹⁸ (49,297 peptide–protein complex structures). For ChEMBL and CycPeptMPDB, deposited HELM

annotations were used directly, so there was no analogous sequence-to-HELM conversion-failure step for these sources.

For Propedia, which provides one-letter amino-acid sequences rather than deposited HELM annotations, we filtered out 18,113 entries (36.7%) containing unknown residues (“X”) or nonstandard amino acids that could not be automatically converted to HELM, resulting in 31,184 peptide–protein pairs. Peptide sequences composed only of standard residues were converted to HELM notation through the standard monomer dictionary and to SMILES using RDKit (version 2025.09.3).¹⁹

We deduplicated peptide sequences within and across data sets based on canonical SMILES, with priority ordering CycPeptMPDB > Propedia > ChEMBL. The final all-source pretraining corpus contained **39,079 unique peptide sequences**: 21,879 from ChEMBL (56.0%), 9,212 from Propedia (23.6%), and 7,988 from CycPeptMPDB (20.4%). This all-source corpus was used for the reference HELM-BERT checkpoint; leave-one-source variants were trained only as diagnostic data-ablation controls.

Of the 39,079 pretraining sequences, 35.4% contained macrocyclic or cross-linked topology (18.0% cyclic, 17.3% lariat), determined from the HELM connection section, and 65.7% contained at least one nonstandard monomer, identified by bracket notation in HELM. The downstream benchmarks were selected to span both canonical and noncanonical peptide regimes; their nonstandard-monomer composition is quantified below.

In HELM syntax, PEPTIDE n uses the integer n as a chain index (PEPTIDE1 for a single-chain peptide, PEPTIDE2 for the second chain of a dimer, and so on). Before tokenization, we applied the same polymer-identifier normalization rule to each HELM string, converting single-chain peptide polymers to PEPTIDE1 so that the integer reflects chain count rather than per-entry source identifiers. This prevents the encoder from learning data set-specific serial-number artifacts in place of chemically meaningful chain topology. Connection records in the HELM connection section were additionally sorted deterministically by source monomer index, target monomer index, and R-group handles, so that semantically equivalent inputs yield the same token sequence.

To support tokenization of HELM strings, we constructed a unified monomer library by merging two sources with priority CycPeptMPDB → ChEMBL: CycPeptMPDB (385 raw monomers) and the PEPTIDE polymerType entries of the ChEMBL v35 monomer library (2851 raw monomers). A ChEMBL monomer was skipped when it shared its HELM symbol with a CycPeptMPDB entry; otherwise, two monomers were treated as equivalent when they shared (i) RDKit isomeric canonical SMILES, (ii) R-group attachment count, and (iii) monomer type (backbone vs. terminal), in which case the CycPeptMPDB symbol was kept as the primary identifier and the ChEMBL symbol stored as an alias. Tautomers and E/Z stereoisomers were retained as distinct because their HELM-level identities differ. The resulting library contains 3,095 unique monomer entries and defines the monomer dictionary used for HELM normalization and SMILES conversion; it is distinct from the 78-token compressed-character vocabulary used by HELM-BERT for MLM. It is released with the code.

Pretraining Objective and Optimization

We trained HELM-BERT using MLM with span masking.^{14,20} We masked 15% of tokens, sampling contiguous span lengths from a geometric distribution ($p = 0.2$) and clipping spans to lengths 1–5. Masked tokens were replaced using an 80–10–10 strategy: 80% replaced with [MASK], 10% replaced with a random token, and 10% left unchanged.

Optimization used AdamW with learning rate 1×10^{-4} , weight decay 0.01, cosine annealing, and gradient clipping (max norm 1.0). Training used FP32 precision with batch size 64 and maximum input length 512 tokens; the 39,079-peptide corpus was split 80/20 into training and validation subsets by seeded random permutation (seed 42), and trained for up to 500 epochs with early stopping (patience = 20) based on validation MLM loss, and the checkpoint with lowest validation loss was selected for downstream evaluation.

Table 1. Experimental Configuration for Membrane Permeability Prediction^a

model	setting	head arch	Enc. params	head params	learning rate
HELM-BERT	Full FT	residual MLP	54.2M	1.18M	Enc: 3×10^{-5} /Head: 1×10^{-4}
	Head FT	residual MLP		1.18M	Head: 1×10^{-4}
	Linear	single linear		0.77K	Head: 1×10^{-3}
MoLFormer-XL	Full FT	official MLP	44.4M	1.18M	Enc: 3×10^{-5} /Head: 3×10^{-5}
	Head FT	official MLP		1.18M	Head: 3×10^{-5}
	Linear	single linear		0.77K	Head: 1×10^{-3}
PeptideCLM	Full FT	official MLP	43.0M	0.59M	Enc: 5×10^{-6} /Head: 5×10^{-6}
	Head FT	official MLP		0.59M	Head: 5×10^{-6}
	Linear	single linear		0.77K	Head: 1×10^{-3}
SMILES-HELM-BERT	Full FT	residual MLP	54.2M	1.18M	Enc: 3×10^{-5} /Head: 1×10^{-4}
	Head FT	residual MLP		1.18M	Head: 1×10^{-4}
	Linear	single linear		0.77K	Head: 1×10^{-3}

^aAll encoders are of comparable scale (43–54 M parameters, excluding small tokenizer-embedding differences). HELM-BERT uses a custom three-layer residual MLP head; MoLFormer-XL uses its official three-layer MLP head (approximately 1.18 M parameters each). PeptideCLM uses its official two-layer MLP (0.59 M parameters). SMILES-HELM-BERT uses the same DeBERTa-style encoder and downstream heads as HELM-BERT, with canonical SMILES tokenized by the MoLFormer-XL tokenizer. Linear probing uses identical single linear layers across all models. Optimizers: AdamW for HELM-BERT, SMILES-HELM-BERT, and linear probing; Adam for MoLFormer-XL/PeptideCLM finetuning.

Embedding Quality Analysis

We evaluated embedding quality on physicochemical regression tasks (Log *P*, molecular weight, TPSA) computed with RDKit, and on topology-related structural classification tasks derived from HELM connectivity.

Structure type was defined from the presence and type of intramolecular connections: cyclic peptides have backbone-only cyclization via R1–R2 connections, lariat peptides involve side-chain connections (e.g., via R3), and linear peptides have no intramolecular connections. The number of connections was defined as the count of intramolecular covalent linkage annotations, capturing ring closures and covalent cross-links as a measure of structural topology.

We evaluated regression and classification via linear probing with 5-fold cross-validation, using the mean-pooled final-layer token representation. Classification used scikit-learn LogisticRegression (*C* = 1.0, lbfgs solver) and regression used Ridge (α = 1.0), both with fixed hyperparameters (no grid search). Additionally, we evaluated *k*-nearest neighbor (*k* = 3) classification and clustering separability metrics (Silhouette,²¹ Davies–Bouldin,²² Caliński–Harabasz²³) on full-dimensional embeddings.

Downstream Task: Membrane Permeability Prediction

We used the deduplicated CycPeptMPDB subset (7,988 entries) and removed 273 outliers with log *P*_{app} ≤ −10.0,¹⁷ yielding 7,715 samples. We performed 10-fold cross-validation; for each fold, 10% was held out as test and the remaining 90% was split into train/validation (80%/10%).

We evaluated two complementary fold-assignment strategies. The *random split* assigns each peptide to a fold by a seeded random permutation. The *Murcko scaffold split* groups peptides by their Bemis–Murcko scaffold²⁴ and assigns each scaffold group as a whole to a single fold, so that no Murcko scaffold is shared between training, validation, and test folds within any round. Scaffolds were computed with RDKit's MurckoScaffold.MurckoScaffoldSmiles (includeChirality = False); peptides for which scaffold extraction failed (e.g., fully disconnected SMILES after conversion) were placed into a single auxiliary group (__no_scaffold__) and assigned together, rather than discarded, so that the number of evaluated peptides matches the random split. Scaffold groups were then sorted in decreasing order of group size (ties broken by the smallest original index) and distributed across *N*_{folds} = 10 folds using a zigzag schedule: for the *i*-th sorted group, the target fold index is $\text{idx}_i = \min(r_i, 2N_{\text{folds}} - r_i - 1)$ with $r_i = i \bmod 2N_{\text{folds}}$, which produces fold sizes within a narrow band while keeping all members of a scaffold group in the same fold. For both splits we used the same 10-round rotation: at round *i*, fold *i* is the test fold, fold $(i + 1) \bmod N_{\text{folds}}$ is the

validation fold, and the remaining folds form the training set (seed 42).

We evaluated three training protocols: full finetuning (encoder + head), head finetuning (encoder frozen), and linear probing (encoder frozen, linear head). The prediction head for HELM-BERT was a three-layer residual MLP; baseline heads followed the official architectures where available. For HELM-BERT, finetuning minimized MSE loss on the raw log *P*_{app} target (no additional normalization), with batch size 32 and up to 200 epochs, using early stopping (patience 20) on validation MSE (seed 42). For MoLFormer-XL and PeptideCLM, we adopted the training configuration (optimizer, batch size, schedule) of each model's original publication without modification, changing only the downstream head to match Table 1.

Downstream Task: Peptide–Protein Interaction Prediction

We constructed a PPI data set from Propedia-derived complexes with negative sampling at a 1:4 positive-to-negative ratio. We used a dual-encoder framework following ChemGLaM²⁵ (Figure S1): a peptide encoder (HELM-BERT or baselines) and a protein encoder (ESM-2,⁸ frozen), whose representations were concatenated and passed to a classification head.

We evaluated two splitting strategies with 5-fold cross-validation (seed 42), both using a 1:4 positive-to-negative sampling ratio within each fold. The *random split* used pair-grouped splitting with no pair overlap across folds (20,057 unique positive pairs). The *cluster-based split* used *k*-means clustering (*k* = 100, seed 42) on aCSM-ALL signatures.^{18,26} These signatures are graph-based descriptors encoding pairwise atomic contact patterns across pharmacophoric atom types; we reduced them to 50 dimensions via PCA and assigned clusters to folds via constrained *k*-means (*k* = 5, ≤ 15% deviation, seed 42). Proteins appearing in multiple splits were assigned to the majority split (ties resolved by prioritizing test > validation > training), and pairs violating fold assignments were removed, yielding 20,055 unique positive pairs. We used binary cross-entropy with positive class weight 4.0 to address class imbalance. All PPI models were trained with batch size 32 and up to 200 epochs, using early stopping (patience 20) on validation loss.

ChEMBL PPI Benchmark Construction. As a second, larger peptide–protein interaction benchmark, we constructed a data set from ChEMBL assay records. We restricted the input to HELM-annotated peptides paired with protein targets that have at least one reported binding activity value in ChEMBL. A pair was labeled *positive* if the reported binding potency was ≤ 1 μM (binary indicator on the curated record; no intermediate potency bucket was used). All other peptide–target pairs within the retained peptide × target pool were used as candidate negatives. Negatives were then drawn *per split* at

fixed 1:4 positive-to-negative ratio from the peptide $\times$ target pool of that split only, so that no negative pair crosses fold boundaries.

We evaluated two splitting strategies with 5-fold cross-validation (seed 42), matching the protocol used for Propedia: a *random split* and a *family based split*. The family split groups pairs by the protein target's family annotation, using the full hierarchical class path from ChEMBL's `protein_class_desc` field as the grouping key; targets lacking class annotation were pooled into a single unclassified group. Because a small number of families contain a disproportionate fraction of the total positive pairs, any family whose size exceeded $1/K$ of all positives (with $K = 5$ folds) was further subdivided into per-target groups, so that no single group alone exceeds one fold's nominal size. Groups were then partitioned across folds using a constrained balanced-partitioning procedure analogous to the aCSM cluster assignment used for Propedia: groups are first assigned greedily to the fold with the smallest current size, and the assignment is then refined by up to 100 iterations of group swaps that reduce the maximum deviation of fold sizes from the nominal $1/K$ fraction, subject to $a \leq 15\%$ per-fold deviation constraint. Within each training/validation split, the validation set was carved out as a further 12.5% of the (training + validation) positive pairs, assembled by accumulating whole groups in decreasing order of size until the target fraction was reached. Negatives were sampled independently inside each train/validation/test split at the same 1:4 ratio.

To quantify the chemical regimes represented by the downstream benchmarks, we computed nonstandard-monomer content at the peptide, pair (for PPI benchmarks), and monomer-token levels. Propedia v2 PPI contains no nonstandard monomers among 9,212 unique peptides, because unknown and nonstandard residues are removed during preprocessing; after this filtering it consists of linear peptides composed of standard amino acids and therefore probes canonical sequence- and structure-level generalization. In the ChEMBL PPI benchmark, 88.9% of 6,334 unique peptides contain at least one nonstandard monomer, and 89.0–89.5% of evaluation pairs involve such a peptide across splits; at the monomer-token level, 21.0% (15,813/75,335) of all occurrences are nonstandard. In ChEMBL PPI, pairs in the positive and negative subsets contain at least one peptide with a nonstandard monomer at comparable rates (90.0% vs. 88.8–89.4%), indicating that the high peptide-level nonstandard composition reflects the benchmark chemistry rather than a class-specific pattern. CycPeptMPDB is the densest modified benchmark: 99.8% of 7,715 unique cyclic peptides contain at least one nonstandard monomer, and 65.5% (41,989/64,144) of monomer-token occurrences are nonstandard, with a mean per-peptide nonstandard fraction of 65.3% (median 66.7%). Thus, the downstream evaluation spans complementary regimes: canonical-only PPI generalization (Propedia), PPI in which nonstandard monomers are widespread at the peptide level but moderate in token-level incorporation (ChEMBL PPI), and dense-modification cyclic-peptide regression (CycPeptMPDB).

Baselines and Experimental Configurations

For SMILES-based baselines, we used MoLFormer–XL⁵ and PeptideCLM.²⁷ Because full MoLFormer–XL weights were not publicly available, we used the official checkpoint pretrained on 10% of the full data set. To isolate the contribution of input representation under an architecture-matched control, we additionally pretrained a SMILES-HELM-BERT baseline from scratch on canonical SMILES strings for the same 39,079-peptide pretraining corpus used for HELM-BERT. SMILES-HELM-BERT uses the same DeBERTa-style encoder architecture as HELM-BERT (six layers, $H = 768$, $A = 12$, with nGiE, disentangled attention, and EMD), together with the same span-masked MLM objective, optimizer, and early stopping criterion (see Table 1). Its tokenizer is the pretrained MoLFormer–XL SMILES tokenizer,⁵ based on the regex-based SMILES tokenization scheme of Schwaller et al.,²⁸ without modification. Thus, SMILES-HELM-BERT differs from HELM-BERT only in input notation and tokenization (canonical SMILES with the MoLFormer–XL tokenizer vs. HELM with the HELM-GPT-style tokenizer), while architecture,

pretraining corpus, objective, optimizer, and training schedule are matched. This baseline is used as a controlled atom-level SMILES comparison rather than as a direct benchmark for peptide-specific SMILES variants. For protein embeddings in PPI prediction, we used ESM-2 variants (35M, 150M, 650M).⁸ As a non-neural baseline, we computed 102 sequence-based descriptors using the peptides Python library,²⁹ including BLOSUM indices, Z-scales, Kidera factors, FASGAI vectors, and physicochemical properties.

Tables 1 and 2 summarize model sizes, head architectures, and learning rates for permeability and PPI experiments, respectively.

Table 2. Experimental Configuration for PPI Prediction^a

peptide encoder	setting	Pep. Enc. Params	concat Dim	head params	head LR
HELM-BERT	Head FT	54.2M	2048	8.4M	1×10^{-4}
	Linear			2.0K	1×10^{-3}
MoLFormer–XL	Head FT	44.4M	2048	8.4M	1×10^{-4}
	Linear			2.0K	1×10^{-3}
PeptideCLM	Head FT	43.0M	2048	8.4M	1×10^{-4}
	Linear			2.0K	1×10^{-3}
SMILES-HELM-BERT	Head FT	54.2M	2048	8.4M	1×10^{-4}
	Linear			2.0K	1×10^{-3}
ESM-2 (650M)	Head FT	651M	2560	13.1M	1×10^{-4}
	Linear			2.6K	1×10^{-3}
ESM-2 (150M)	Head FT	148M	1920	7.4M	1×10^{-4}
	Linear			1.9K	1×10^{-3}
ESM-2 (35M)	Head FT	34M	1760	6.6M	1×10^{-4}
	Linear			1.8K	1×10^{-3}
Peptide descriptors	Head FT	–	1382	3.8M	1×10^{-4}
	Linear			1.4K	1×10^{-3}

^aAll peptide encoders are paired with a frozen ESM-2 (650M) as the protein encoder. Peptide and protein representations are concatenated before prediction. Head finetuning uses a unified three-layer residual MLP; linear probing uses a single linear layer. SMILES-HELM-BERT uses the same peptide encoder/head dimensions as HELM-BERT but receives canonical SMILES tokens from the MoLFormer–XL tokenizer. Head size scales with input dimension ($D_{\text{pep}} + D_{\text{prot}}$). Optimizer: AdamW.

Statistical Comparison Framework

For each downstream task and evaluation setting, we performed pairwise model comparisons on fold-wise performance values using the corrected resampled *t*-test of Nadeau and Bengio,³⁰ which augments the standard paired *t*-test with a $(1/k + 1/(k - 1))$ correction factor to compensate for the nonindependence of overlapping training sets in *k*-fold cross-validation. Raw two-sided *p*-values were then adjusted using the Benjamini–Hochberg false discovery rate procedure,³¹ applied across all baseline/variant $\times$ metric comparisons within each table or table panel (panel-wide correction). This panel-wide convention is used uniformly throughout both the main-text tables and the Supporting Information; in the main-text tables, footnote markers indicate cells that are FDR-significantly different from HELM-BERT for the displayed metric cell, while the narrative emphasizes the primary metric (R^2 for permeability and ROC-AUC for PPI). A comparison was declared *FDR-significant* when $p_{\text{adj}} < 0.05$; values in the range $0.05 \leq p_{\text{adj}} < 0.10$ are reported as *marginal*. Alongside adjusted *p*-values, we report the paired Cohen's *d* effect size³² using conventional magnitude thresholds: $|d| < 0.2$ (negligible), $0.2 \leq |d| < 0.5$ (small), $0.5 \leq |d| <$

0.8 (medium), and $|d| \geq 0.8$ (large). For the permeability task we used $k = 10$ folds; for the PPI task $k = 5$.

RESULTS

Pretraining an Encoder-Only Transformer Directly on HELM Notation

HELM-BERT is an encoder-only transformer pretrained on a MLM objective over HELM notation. The input is tokenized with the compressed-character HELM tokenizer described above, and includes both simple-polymer and connectivity descriptors. The model architecture is based on DeBERTa-style disentangled attention¹⁴ with additional nGiE and an EMD module (Figure 1), designed to strengthen relative-position modeling and improve masked-token reconstruction for complex sequences.

We pretrained HELM-BERT for 104 epochs using FP32 precision with a batch size of 64 on a single NVIDIA GH200 (Grace Hopper, 96 GB HBM3; ~57 GB peak memory used) on the Miyabi supercomputer. Training required approximately 2.6 h (159 min), and the lowest validation loss was 0.3548 at epoch 84 (Figure 2). The final pretrained checkpoint

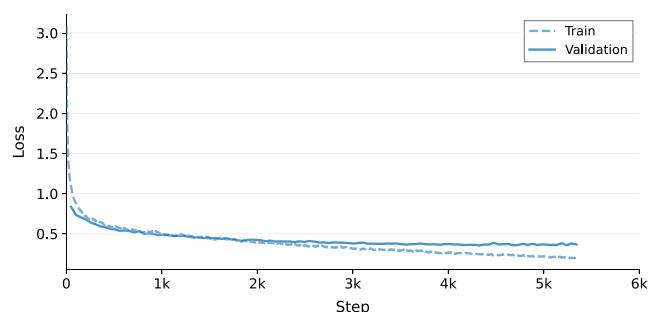


Figure 2. Pretraining loss curves. Training and validation of MLM loss over the course of pretraining. The model was trained for 104 epochs with early stopping (patience = 20).

was selected based on minimum validation MLM loss and used for all downstream evaluations. All downstream finetuning, linear probing, and embedding evaluations were performed using the Miyabi supercomputer (NVIDIA GH200) and a local NVIDIA RTX PRO 6000 Blackwell workstation.

Topology-Aware HELM Representations Improve Cyclic Peptide Membrane Permeability Prediction

We evaluated HELM-BERT on cyclic peptide membrane permeability prediction using CycPeptMPDB, a data set containing experimental permeability measurements for cyclic peptides with diverse chemistries and cyclization patterns. Because permeability in cyclic peptides is strongly affected by macrocyclization and modification patterns, this task provides a direct test of whether topology-aware representations confer predictive advantages. To characterize the evaluation regimes, we first visualized the CycPeptMPDB data set under both a random 10-fold split and a Murcko scaffold-based 10-fold split using t-SNE projections of molecular descriptors (Figure 3). The random split mixes structurally related peptides across folds, whereas the scaffold split enforces that all peptides sharing a Murcko scaffold are assigned to the same fold, producing spatially distinct fold assignments and thus a more stringent test of out-of-distribution generalization.

HELM-BERT achieved the best mean performance across all three training protocols on the random split (Table 3).

Under full finetuning, HELM-BERT reached $R^2 = 0.668 \pm 0.050$, significantly exceeding the two external pretrained SMILES-based peptide encoders MoLFormer-XL ($R^2 = 0.578 \pm 0.043$; $d = 1.97$, $p_{\text{adj}} = 0.0035$) and PeptideCLM ($R^2 = 0.536 \pm 0.025$; $d = 3.50$, $p_{\text{adj}} = 1.3 \times 10^{-4}$). The gap to the architecture-matched SMILES-HELM-BERT control ($R^2 = 0.646 \pm 0.038$) was small and not statistically significant after FDR correction ($d = 0.74$, $p_{\text{adj}} = 0.155$). Because SMILES-HELM-BERT uses the same DeBERTa-style architecture, corpus, MLM objective, optimizer, and training schedule as HELM-BERT, this comparison provides a stricter estimate of the remaining contribution of HELM notation and tokenization; on its own, it does not establish an FDR-significant HELM-vs-SMILES advantage under full finetuning. Under head finetuning, HELM-BERT significantly exceeded all three baselines, including the architecture-matched SMILES-HELM-BERT control ($d = 1.33$, $p_{\text{adj}} = 0.0187$). Under linear probing, HELM-BERT significantly outperformed MoLFormer-XL ($d = 2.46$) and SMILES-HELM-BERT ($d = 2.48$, both $p_{\text{adj}} = 0.0010$), with the gap to PeptideCLM marginal after FDR correction ($d = 0.85$, $p_{\text{adj}} = 0.108$). Per-comparison statistics and fold-wise results are reported in Tables S2–S5.

Linear probing provides the most direct comparison of representation quality, as it isolates the information already present in frozen embeddings. In this setting, HELM-BERT significantly exceeded MoLFormer-XL ($\Delta R^2 = 0.087$) and the architecture-matched SMILES-HELM-BERT control ($\Delta R^2 = 0.069$), while the gap to PeptideCLM was marginal ($\Delta R^2 = 0.034$, $p_{\text{adj}} = 0.108$). SMILES-HELM-BERT underperformed HELM-BERT under linear probing but not under full finetuning. This pattern indicates that HELM notation makes permeability-relevant structural features more linearly accessible in the frozen representation, even when full finetuning can partially compensate for atom-level input encoding.

To assess generalization under distribution shift, we additionally evaluated all four models on a Murcko scaffold split (Table 4). As expected, absolute performance decreased relative to the random split. Nevertheless, HELM-BERT achieved the best mean value for every metric under all three training protocols. Under full finetuning, HELM-BERT significantly outperformed MoLFormer-XL ($\Delta R^2 = 0.109$, $d = 1.30$, $p_{\text{adj}} = 0.0335$) and PeptideCLM ($\Delta R^2 = 0.244$, $d = 3.18$, $p_{\text{adj}} = 4.2 \times 10^{-4}$), while the positive gap to SMILES-HELM-BERT was not FDR-significant ($\Delta R^2 = 0.035$, $d = 0.74$, $p_{\text{adj}} = 0.154$). Under head finetuning, HELM-BERT remained the top point estimate, but no R^2 comparison reached FDR significance: the gap to MoLFormer-XL was negligible ($\Delta R^2 = 0.004$), whereas the larger mean gaps to PeptideCLM and SMILES-HELM-BERT did not reach significance because of the high fold-level variance characteristic of scaffold splits. Under linear probing, HELM-BERT again achieved the highest mean values and significantly outperformed SMILES-HELM-BERT ($\Delta R^2 = 0.143$, $d = 1.75$, $p_{\text{adj}} = 0.0124$), while the gaps to MoLFormer-XL and PeptideCLM were positive but not FDR-significant. Overall, the scaffold split supports HELM-BERT's top-ranked mean performance and robustness under scaffold-level distribution shift, while showing that statistical separation is strongest under full finetuning and limited by fold-level variance in the frozen-encoder settings. Full scaffold-split statistics are included in Tables S2–S4.

Importantly, we observed that the advantage is not contingent on potential benchmark overlap: excluding benchmark-derived peptides from pretraining did not materially

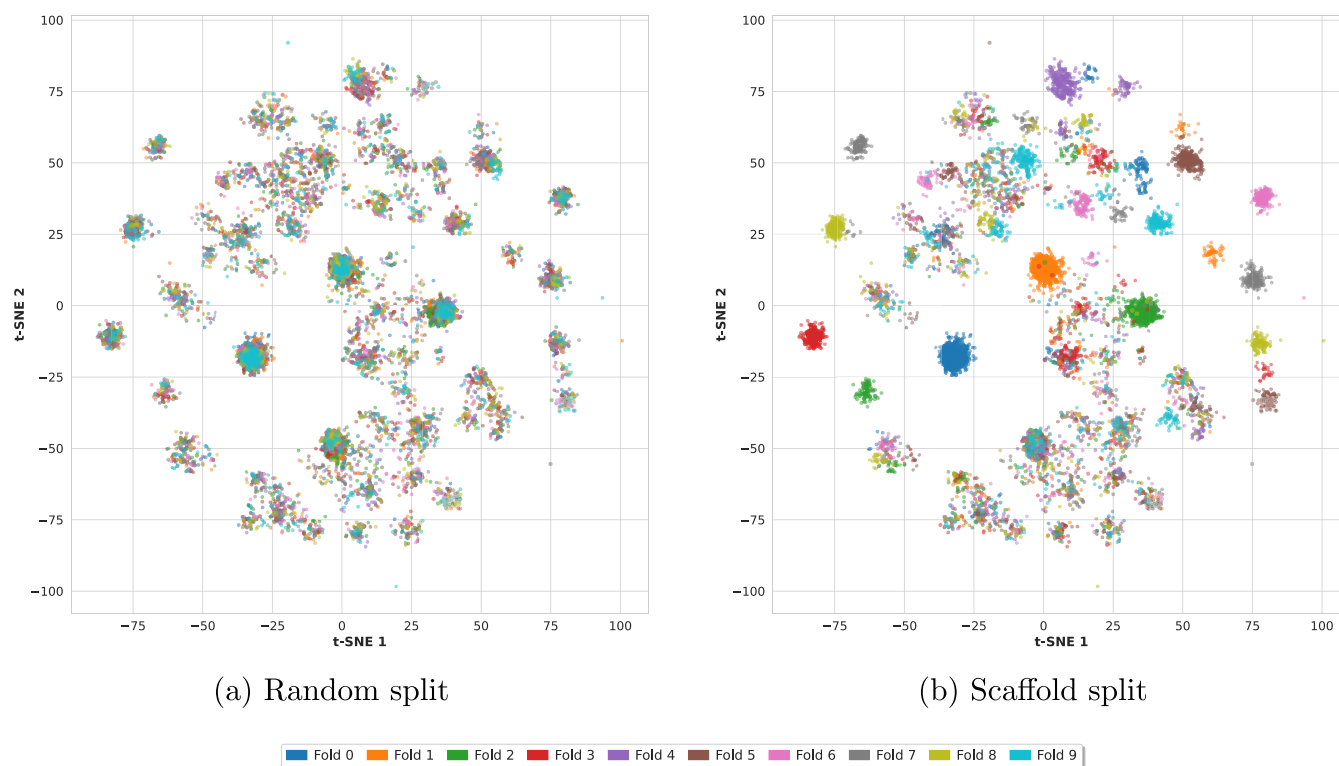


Figure 3. t-SNE projections of CycPeptMPDB data set splits. (a) Random 10-fold split and (b) Murcko scaffold-based 10-fold split. Colors indicate fold assignment. Under the scaffold split, peptides sharing the same Murcko scaffold are kept in the same fold, yielding more spatially distinct fold regions than the random split and a more stringent generalization test.

Table 3. Performance Comparison on the CycPeptMPDB Permeability Dataset (Random Split)^b

model	$R^2 \uparrow$	pearson $r \uparrow$	RMSE $\downarrow$	MAE $\downarrow$
Full Finetuning				
HELM-BERT	0.6683 ± 0.0497	0.8220 ± 0.0282	0.4504 ± 0.0279	0.3147 ± 0.0156
MolFormer-XL	0.5776 ± 0.0434^a	0.7673 ± 0.0247^a	0.5094 ± 0.0267^a	0.3668 ± 0.0167^a
PeptideCLM	0.5360 ± 0.0245^a	0.7413 ± 0.0127^a	0.5344 ± 0.0158^a	0.3847 ± 0.0109^a
SMILES-HELM-BERT	<u>0.6461 ± 0.0380</u>	<u>0.8103 ± 0.0214</u>	<u>0.4660 ± 0.0203</u>	<u>0.3259 ± 0.0113</u>
Head Finetuning				
HELM-BERT	0.6416 ± 0.0369	0.8054 ± 0.0200	0.4690 ± 0.0210	0.3366 ± 0.0135
MolFormer-XL	0.5510 ± 0.0348^a	0.7446 ± 0.0218^a	0.5255 ± 0.0224^a	0.3914 ± 0.0130^a
PeptideCLM	0.4297 ± 0.0256^a	0.6569 ± 0.0193^a	0.5927 ± 0.0245^a	0.4426 ± 0.0141^a
SMILES-HELM-BERT	<u>0.5942 ± 0.0297^a</u>	<u>0.7758 ± 0.0177^a</u>	<u>0.4994 ± 0.0135^a</u>	<u>0.3673 ± 0.0098^a</u>
Linear Probing				
HELM-BERT	0.3936 ± 0.0248	0.6353 ± 0.0133	0.6112 ± 0.0228	0.4578 ± 0.0178
MolFormer-XL	0.3070 ± 0.0244^a	0.5611 ± 0.0219^a	0.6535 ± 0.0256^a	0.4950 ± 0.0145^a
PeptideCLM	<u>0.3597 ± 0.0213</u>	<u>0.6035 ± 0.0185^a</u>	<u>0.6282 ± 0.0246</u>	<u>0.4703 ± 0.0152</u>
SMILES-HELM-BERT	0.3245 ± 0.0218^a	0.5757 ± 0.0166^a	0.6451 ± 0.0220^a	0.4876 ± 0.0180^a

^aSignificantly different from HELM-BERT on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 10$). Metrics: coefficient of determination (R^2), Pearson correlation (r), root mean squared error (RMSE), mean absolute error (MAE). SMILES-HELM-BERT is an architecture-matched DeBERTa-style encoder pretrained on canonical SMILES tokens with the MolFormer-XL tokenizer instead of HELM notation with the HELM-GPT-style tokenizer, included as an input-representation control. ^bBest results are **bolded**, second-best are underlined. $\uparrow$: higher is better, $\downarrow$: lower is better.

degrade downstream permeability performance (Table 6), supporting the interpretation that HELM-BERT learns transferable features rather than memorizing benchmark instances.

Architectural Components Contribute Jointly to Permeability

To identify which architectural components are responsible for HELM-BERT's gains, we performed ablation studies on the encoder architecture and pretraining objective (Table 5).

Among single-component ablations, removing any one of disentangled attention, EMD, nGiE, span masking, or connectivity tokens produced only modest mean drops in downstream R^2 under the random split (each $\Delta R^2 \leq 0.019$; all $p_{\text{adj}} > 0.20$ using the FDR-adjusted comparisons reported in Table S6). Replacing the full DeBERTa-style architecture/objective stack with a vanilla transformer encoder trained by token-level MLM ("Vanilla-BERT") was the only architectural ablation that reached FDR significance under the random split

Table 4. Performance Comparison on the CycPeptMPDB Permeability Dataset (Murcko Scaffold Split, Evaluating Generalization under Distribution Shift)^b

model	$R^2 \uparrow$	pearson $r \uparrow$	RMSE $\downarrow$	MAE $\downarrow$
Full Finetuning				
HELM-BERT	0.4833 $\pm$ 0.1014	0.7247 $\pm$ 0.0580	0.5632 $\pm$ 0.0735	0.3999 $\pm$ 0.0423
MoLFormer-XL	0.3741 $\pm$ 0.0997 ^a	0.6371 $\pm$ 0.0557 ^a	0.6196 $\pm$ 0.0578 ^a	0.4539 $\pm$ 0.0355 ^a
PeptideCLM	0.2389 $\pm$ 0.1363 ^a	0.5555 $\pm$ 0.0634 ^a	0.6855 $\pm$ 0.0945 ^a	0.5051 $\pm$ 0.0717 ^a
SMILES-HELM-BERT	<u>0.4486 $\pm$ 0.0914</u>	<u>0.7002 $\pm$ 0.0484</u>	<u>0.5835 $\pm$ 0.0770</u>	<u>0.4173 $\pm$ 0.0448</u>
Head Finetuning				
HELM-BERT	0.3482 $\pm$ 0.1118	0.6277 $\pm$ 0.0616	0.6344 $\pm$ 0.0856	0.4704 $\pm$ 0.0627
MoLFormer-XL	<u>0.3446 $\pm$ 0.1030</u>	<u>0.6002 $\pm$ 0.0707</u>	<u>0.6364 $\pm$ 0.0814</u>	<u>0.4715 $\pm$ 0.0523</u>
PeptideCLM	0.1803 $\pm$ 0.0874	0.4675 $\pm$ 0.0636	0.7114 $\pm$ 0.0688	0.5341 $\pm$ 0.0483
SMILES-HELM-BERT	0.2082 $\pm$ 0.1127	0.5162 $\pm$ 0.0904	0.7007 $\pm$ 0.0969	0.5183 $\pm$ 0.0631
Linear Probing				
HELM-BERT	0.2226 $\pm$ 0.1175	0.5411 $\pm$ 0.0720	0.6911 $\pm$ 0.0640	0.5328 $\pm$ 0.0542
MoLFormer-XL	<u>0.1655 $\pm$ 0.0864</u>	<u>0.4410 $\pm$ 0.0812</u>	<u>0.7175 $\pm$ 0.0640</u>	0.5456 $\pm$ 0.0531
PeptideCLM	0.1621 $\pm$ 0.0740	0.4264 $\pm$ 0.0741	0.7198 $\pm$ 0.0702	<u>0.5430 $\pm$ 0.0547</u>
SMILES-HELM-BERT	0.0794 $\pm$ 0.1807 ^a	0.4263 $\pm$ 0.0756 ^a	0.7519 $\pm$ 0.0877 ^a	0.5757 $\pm$ 0.0744 ^a

^aSignificantly different from HELM-BERT on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 10$). Metrics: coefficient of determination (R^2), Pearson correlation (r), root mean squared error (RMSE), mean absolute error (MAE). SMILES-HELM-BERT is the architecture-matched DeBERTa-style SMILES control defined in Table 3. Nonsignificant differences of large magnitude (e.g., $\Delta R^2 = 0.14$ – 0.17 , $d = 0.76$ – 1.14 in head finetuning) reflect the high fold-level variance characteristic of scaffold splits with $n = 10$ folds. ^bMurcko-scaffold split: peptides sharing the same Murcko scaffold are assigned to the same fold so that all test peptides have scaffolds unseen in training. Best results are **bolded**, second-best are underlined. $\uparrow$: higher is better, $\downarrow$: lower is better.

Table 5. Ablation Study on Architecture and Pretraining Objective (CycPeptMPDB, Full Finetuning)^b

Variant	$R^2 \uparrow$	Pearson $r \uparrow$	RMSE $\downarrow$	MAE $\downarrow$
Random Split				
HELM-BERT (full)	0.6683 $\pm$ 0.0497	0.8220 $\pm$ 0.0282	0.4504 $\pm$ 0.0279	0.3147 $\pm$ 0.0156
w/o Disentangled attention	0.6519 $\pm$ 0.0438	0.8144 $\pm$ 0.0237	0.4621 $\pm$ 0.0277	0.3238 $\pm$ 0.0103
w/o EMD	0.6573 $\pm$ 0.0504	0.8152 $\pm$ 0.0279	0.4579 $\pm$ 0.0308	0.3185 $\pm$ 0.0190
w/o nGiE	0.6499 $\pm$ 0.0527	0.8105 $\pm$ 0.0334	0.4628 $\pm$ 0.0299	0.3247 $\pm$ 0.0184
w/o Span masking	0.6511 $\pm$ 0.0554	0.8147 $\pm$ 0.0308	0.4616 $\pm$ 0.0337	0.3175 $\pm$ 0.0166
w/o Connectivity tokens	<u>0.6642 $\pm$ 0.0504</u>	<u>0.8191 $\pm$ 0.0296</u>	<u>0.4531 $\pm$ 0.0302</u>	0.3110 $\pm$ 0.0171
Vanilla-BERT	0.6270 $\pm$ 0.0439 ^a	0.7972 $\pm$ 0.0256 ^a	0.4782 $\pm$ 0.0240 ^a	0.3328 $\pm$ 0.0174 ^a
Scaffold Split				
HELM-BERT (full)	0.4833 $\pm$ 0.1014	0.7247 $\pm$ 0.0580	0.5632 $\pm$ 0.0735	0.3999 $\pm$ 0.0423
w/o Disentangled attention	0.4652 $\pm$ 0.0926	0.7131 $\pm$ 0.0570	0.5741 $\pm$ 0.0740	0.4085 $\pm$ 0.0413
w/o EMD	0.4432 $\pm$ 0.1202	0.6990 $\pm$ 0.0548	0.5853 $\pm$ 0.0901	0.4165 $\pm$ 0.0572
w/o nGiE	0.4886 $\pm$ 0.1230	<u>0.7239 $\pm$ 0.0695</u>	0.5600 $\pm$ 0.0904	0.3954 $\pm$ 0.0542
w/o Span masking	0.4746 $\pm$ 0.0719	0.7171 $\pm$ 0.0474	0.5698 $\pm$ 0.0700	0.4022 $\pm$ 0.0383
w/o Connectivity tokens	0.4192 $\pm$ 0.1233	0.6791 $\pm$ 0.0779	0.5978 $\pm$ 0.0909	0.4268 $\pm$ 0.0628
Vanilla-BERT	0.4437 $\pm$ 0.0817	0.6893 $\pm$ 0.0537	0.5862 $\pm$ 0.0751	0.4195 $\pm$ 0.0473

^aSignificantly different from HELM-BERT (full) on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 10$). Metrics: coefficient of determination (R^2), Pearson correlation (r), root mean squared error (RMSE), mean absolute error (MAE). ^bAll variants are pretrained on the full corpus and fine-tuned identically on CycPeptMPDB under full finetuning. “Vanilla-BERT” denotes a standard transformer encoder without disentangled attention, nGiE, or EMD, trained with token-level MLM instead of span masking. “w/o Connectivity tokens” removes explicit HELM connectivity descriptors by replacing the connection list with a placeholder token at the input. Best results within each panel are **bolded**, second-best are underlined. $\uparrow$: higher is better, $\downarrow$: lower is better.

($\Delta R^2 = 0.041$, $d = 1.85$, $p_{\text{adj}} = 0.018$). This pattern indicates that the architectural components contribute jointly to downstream performance rather than through a single dominant mechanism. Under the more stringent Murcko scaffold split (Table 5), no architectural ablation, including the cumulative Vanilla-BERT variant, reached FDR significance ($|d| \leq 0.59$ on R^2 , all $p_{\text{adj}} > 0.69$). The scaffold results were retained in the main table and reconciled with the full scaffold permeability benchmark (Table 4); the higher fold-level variance under scaffold partitioning limits single-component attribution in this regime.

Pretraining Benefits are Substantial and Robust to Corpus Composition

We next assessed whether performance gains depended on the pretraining corpus composition (Table 6; full statistics in Table S7). Under the random split, the from-scratch baseline underperformed the pretrained model ($R^2 = 0.616 \pm 0.050$ vs. 0.668 ± 0.050 ; $\Delta R^2 = 0.052$, $d = 2.37$, $p_{\text{adj}} = 0.002$), confirming the benefit of MLM pretraining on the upstream HELM corpus. Excluding any single source (ChEMBL, Propedia, or CycPeptMPDB) from pretraining did not produce a statistically significant change in downstream R^2 after FDR correction (all $d \leq 0.60$, $p_{\text{adj}} \geq 0.45$). Under the

Table 6. Ablation Study on Pretraining Data Composition (CycPeptMPDB, Full Finetuning)^b

variant	$R^2 \uparrow$	pearson $r \uparrow$	RMSE $\downarrow$	MAE $\downarrow$
Random Split				
HELM-BERT (full)	0.6683 $\pm$ 0.0497	0.8220 $\pm$ 0.0282	0.4504 $\pm$ 0.0279	0.3147 $\pm$ 0.0156
w/o ChEMBL	0.6608 $\pm$ 0.0602	<u>0.8187 $\pm$ 0.0326</u>	0.4549 $\pm$ 0.0402	0.3145 $\pm$ 0.0236
w/o Propedia	<u>0.6614 $\pm$ 0.0583</u>	0.8178 $\pm$ 0.0334	<u>0.4548 $\pm$ 0.0369</u>	0.3175 $\pm$ 0.0236
w/o CycPeptMPDB	0.6489 $\pm$ 0.0510	0.8112 $\pm$ 0.0284	0.4636 $\pm$ 0.0318	0.3192 $\pm$ 0.0178
From scratch	0.6161 $\pm$ 0.0500 ^a	0.7897 $\pm$ 0.0275 ^a	0.4850 $\pm$ 0.0265 ^a	0.3430 $\pm$ 0.0143 ^a
Scaffold Split				
HELM-BERT (full)	0.4833 $\pm$ 0.1014	0.7247 $\pm$ 0.0580	0.5632 $\pm$ 0.0735	0.3999 $\pm$ 0.0423
w/o ChEMBL	0.4531 $\pm$ 0.1210	0.7035 $\pm$ 0.0551	0.5793 $\pm$ 0.0839	0.4159 $\pm$ 0.0531
w/o Propedia	0.4561 $\pm$ 0.0753	<u>0.7035 $\pm$ 0.0474</u>	0.5796 $\pm$ 0.0708	0.4096 $\pm$ 0.0392
w/o CycPeptMPDB	<u>0.4567 $\pm$ 0.0968</u>	0.7002 $\pm$ 0.0574	<u>0.5790 $\pm$ 0.0840</u>	<u>0.4085 $\pm$ 0.0505</u>
From scratch	0.3671 $\pm$ 0.1071 ^a	0.6467 $\pm$ 0.0651 ^a	0.6239 $\pm$ 0.0732 ^a	0.4489 $\pm$ 0.0445 ^a

^aSignificantly different from HELM-BERT (full) on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 10$). ^b“From scratch” denotes training with random initialization and no MLM pretraining on the upstream corpus. “w/o *X*” denotes pretraining after excluding component *X* from the pretraining corpus. Best results within each panel are **bolded**, second-best are underlined. $\uparrow$: higher is better, $\downarrow$: lower is better.

Murcko scaffold split (Table 6), the from-scratch baseline remained FDR-significantly below the pretrained model ($R^2 = 0.367 \pm 0.107$ vs. 0.483 ± 0.101 ; $\Delta R^2 = 0.116$, $d = 1.70$, $p_{\text{adj}} = 0.019$), while each individual source exclusion remained nonsignificant ($|d| \leq 0.40$ on R^2 , all $p_{\text{adj}} \geq 0.47$). In particular, excluding CycPeptMPDB itself from pretraining did not significantly degrade downstream permeability performance under either split, supporting robustness to potential benchmark overlap. This combination of a reliable pretraining gain together with nonsignificant effects of any single corpus component reinforces the conclusion that HELM-BERT’s advantage stems from the pretraining procedure itself rather than from any one data source.

Architectural Components Show Joint and Partially Compensatory Behavior

While the architectural ablations above quantify the *removal* effect of each component on downstream permeability, we additionally examined whether the full architecture/objective stack shows evidence of use during pretraining and whether removing a component induces compensatory changes elsewhere. Combining MLM training curves (Figure 4) with parameter-norm analysis of the pretrained HELM-BERT and its architectural variants (Table S8), we observed three consistent patterns. First, the nGiE convolutional kernel norm grows substantially during pretraining (to 30.17 in HELM-BERT full), indicating that the *n*-gram induced encoding is learned rather than left inert. Second, removing

disentangled attention destabilizes pretraining itself: the variant required 64% more epochs to reach convergence and attained a 70% higher terminal validation loss (Figure 4), and at the parameter level the absolute position embedding norm increased sharply by +123% ($20.56 \rightarrow 45.82$) with the per-position standard deviation tripling ($0.41 \rightarrow 1.33$); this pattern suggests that absolute position embeddings partially compensate for relative-position modeling during MLM pretraining but do not fully substitute for it. Third, when span masking is removed, both the nGiE and position-embedding norms are slightly compressed (28.44 and 18.60), and the Vanilla-BERT variant, where disentangled attention, EMD, and nGiE are all removed, shows the lowest position-embedding norm and per-position variance (16.58 and 0.203), consistent with a simpler architecture/objective stack with fewer structural inductive biases. Together with the ablation results, this component-level analysis supports retaining the full architecture as the reference model: removing individual components did not produce an FDR-significant decrease in downstream permeability, so we do not identify any single component as individually indispensable, and instead interpret the combined architecture/objective stack as jointly contributing. Consistent with this interpretation, the cumulative Vanilla-BERT replacement is the only architectural ablation that reaches FDR significance under the random split.

HELM-BERT Embeddings Preserve Linearly Accessible Topology Information

We also evaluated whether topology information explicitly specified in HELM remains linearly accessible in the final embedding space. Using the pretraining corpus, we assessed embedding quality on physicochemical regression tasks (Log *P*, molecular weight, TPSA) and on topology-related structural classification tasks (structure type and number of connections).

All encoders achieved strong performance on physicochemical property regression ($R^2 > 0.95$), consistent with their ability to capture chemical composition. MolFormer–XL significantly outperformed HELM-BERT on Log *P* ($p_{\text{adj}} = 0.003$, $d = 4.52$), and the SMILES-based baselines outperformed HELM-BERT on molecular weight and TPSA ($p_{\text{adj}} < 0.001$, $d > 9$). This is expected because SMILES represents atom-level connectivity directly, whereas HELM operates at

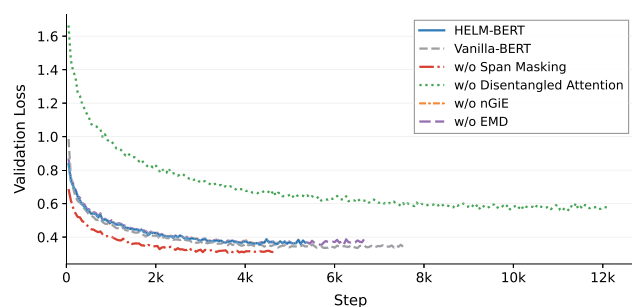


Figure 4. Pretraining MLM loss curves under architectural ablations. Validation loss for HELM-BERT and architectural variants over training epochs.

Table 7. Embedding Quality Evaluation of MLM-Pretrained Encoders Using Full-Dimensional Representations^b

Task	Metric	HELM-BERT	MoLFormer-XL	PeptideCLM
Physicochemical Properties (Regression)				
LogP	$R^2 \uparrow$	<u>0.9535 ± 0.0049</u>	0.9638 ± 0.0031^a	0.9527 ± 0.0018
	MAE ↓	<u>0.98 ± 0.01</u>	0.82 ± 0.01	1.00 ± 0.01
Molecular weight	$R^2 \uparrow$	0.9770 ± 0.0003	<u>0.9842 ± 0.0009^a</u>	0.9900 ± 0.0001^a
	MAE ↓	125.71 ± 0.49	<u>96.25 ± 1.19</u>	82.28 ± 1.19
TPSA	$R^2 \uparrow$	0.9779 ± 0.0005	<u>0.9840 ± 0.0008^a</u>	0.9880 ± 0.0002^a
	MAE ↓	55.03 ± 0.45	<u>43.15 ± 0.60</u>	39.94 ± 0.61
Structural Features (Classification and Separability)				
Structure type	Accuracy (K-NN) ↑	0.9993	<u>0.9926</u>	0.9910
	Linear-classifier accuracy ↑	0.9996 ± 0.0002	<u>0.9810 ± 0.0015^a</u>	0.9763 ± 0.0018 ^a
	Linear-classifier MCC ↑	0.9996	<u>0.9714</u>	0.9649
	Silhouette ↑	0.1060	0.0720	<u>0.0956</u>
	Davies–Bouldin ↓	3.0179	<u>3.1729</u>	3.9966
Number of connections	Caliński–Harabasz ↑	2906	<u>2686</u>	2133
	Accuracy (K-NN) ↑	0.9980	<u>0.9923</u>	0.9911
	Linear-classifier accuracy ↑	0.9975 ± 0.0006	<u>0.9788 ± 0.0016^a</u>	0.9739 ± 0.0027 ^a
	Linear-classifier MCC ↑	0.9979	<u>0.9669</u>	0.9603
	Silhouette ↑	0.0438	<u>−0.0736</u>	−0.0788
	Davies–Bouldin ↓	2.2702	2.9537	<u>2.7056</u>
	Caliński–Harabasz ↑	704	<u>668</u>	558

^aSignificantly different from HELM-BERT on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 10$); statistical tests were applied only to cross-validated metrics (R^2 and linear-classifier accuracy). ^bLinear probing and K-NN classification assess predictive performance; cluster validity indices (applied to ground-truth labels) quantify class separability. Best results are **bolded**, second-best are underlined. ↑: higher is better, ↓: lower is better.

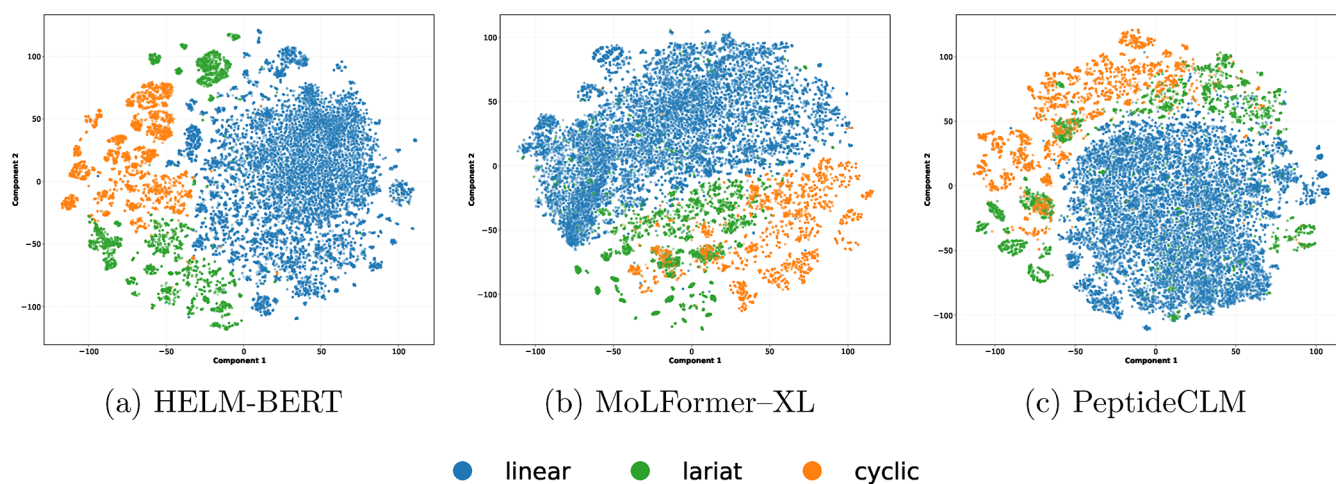


Figure 5. t-SNE projections of pretrained embeddings colored by structure type. (a) HELM-BERT, (b) MoLFormer-XL, and (c) PeptideCLM. Points are colored by peptide structure type (linear, cyclic, lariat); a shared legend is shown below the panels.

the monomer level and abstracts away explicit atomic graph structure.

In contrast, for topology-related structural tasks, HELM-BERT embeddings showed near-perfect linear separability. For structure type classification (linear vs. cyclic vs. lariat), HELM-BERT achieved 0.9996 ± 0.0002 linear-classifier accuracy, exceeding MoLFormer-XL (0.9810 ± 0.0015) and PeptideCLM (0.9763 ± 0.0018) with large effect sizes ($d > 12$, $p_{\text{adj}} < 0.001$; Table S9). Similar gains were observed for number of connections classification ($d > 10$, $p_{\text{adj}} < 0.001$). Clustering metrics (Silhouette, Davies–Bouldin, Caliński–Harabasz) further supported that HELM-BERT produces a more topology-structured embedding space (Table 7). Two-dimensional projections using t-SNE also revealed clearer separation by structural category for HELM-BERT (Figures 5, S2–S7,

and Table S10). Structure type and connection-count labels are derived from HELM connectivity annotations that are also present in the input. Therefore, this analysis should be interpreted as a consistency and accessibility check (that the encoder preserves explicit topology information in the final embedding) rather than as independent evidence of de novo structural understanding.

These results indicate a functional dichotomy: SMILES-based embeddings are strong for atom-level scalar property regression, whereas HELM-BERT embeddings preserve discrete HELM-specified topological features in a linearly decodable form. Given that macrocyclization patterns and cross-links strongly influence permeability and binding modes, explicit topology encoding is consistent with the downstream gains observed in permeability prediction.

Table 8. Performance Comparison Using Propedia PPI Dataset with Random Split^b

model	params	ROC-AUC ↑	PR-AUC ↑	MCC ↑	Bal. Acc ↑
MLP					
HELM-BERT	54.2M	0.9419 ± 0.0070	0.8330 ± 0.0197	0.6700 ± 0.0402	0.8695 ± 0.0109
ESM-2 (650M)	651M	0.9416 ± 0.0055	0.8324 ± 0.0152	0.6704 ± 0.0222	0.8703 ± 0.0089
ESM-2 (150M)	148M	<u>0.9434 ± 0.0031</u>	<u>0.8418 ± 0.0076</u>	<u>0.6822 ± 0.0099</u>	<u>0.8740 ± 0.0037</u>
ESM-2 (35M)	34M	0.9466 ± 0.0055	0.8474 ± 0.0152	0.6886 ± 0.0252	0.8787 ± 0.0073
PeptideCLM	43.0M	0.9190 ± 0.0078	0.7807 ± 0.0204	0.6036 ± 0.0331	0.8405 ± 0.0092
MoLFormer-XL	44.4M	0.9218 ± 0.0073	0.7866 ± 0.0170	0.6178 ± 0.0244	0.8427 ± 0.0134
Peptide descriptors	—	0.9320 ± 0.0038	0.8233 ± 0.0090	0.6589 ± 0.0144	0.8619 ± 0.0061
SMILES-HELM-BERT	54.2M	0.8992 ± 0.0053	0.7244 ± 0.0081	0.5385 ± 0.0252	0.8158 ± 0.0083
w/o Propedia	54.2M	0.9351 ± 0.0025	0.8169 ± 0.0123	0.6505 ± 0.0282	0.8618 ± 0.0050
Linear					
HELM-BERT	54.2M	<u>0.6167 ± 0.0059</u>	0.2750 ± 0.0068	0.1345 ± 0.0071	0.5828 ± 0.0051
ESM-2 (650M)	651M	0.6205 ± 0.0026	0.2706 ± 0.0039	<u>0.1315 ± 0.0059</u>	<u>0.5809 ± 0.0037</u>
ESM-2 (150M)	148M	0.6109 ± 0.0031	0.2626 ± 0.0042 ^a	0.1235 ± 0.0039	0.5757 ± 0.0026
ESM-2 (35M)	34M	0.6047 ± 0.0041	0.2557 ± 0.0053 ^a	0.1176 ± 0.0066	0.5719 ± 0.0038
PeptideCLM	43.0M	0.5956 ± 0.0045 ^a	0.2625 ± 0.0087	0.1066 ± 0.0049 ^a	0.5658 ± 0.0035 ^a
MoLFormer-XL	44.4M	0.5949 ± 0.0047 ^a	0.2615 ± 0.0023	0.0991 ± 0.0061 ^a	0.5609 ± 0.0039 ^a
Peptide descriptors	—	0.5584 ± 0.0041 ^a	0.2311 ± 0.0039 ^a	0.0665 ± 0.0127 ^a	0.5412 ± 0.0081 ^a
SMILES-HELM-BERT	54.2M	0.5783 ± 0.0028 ^a	0.2473 ± 0.0043 ^a	0.0859 ± 0.0035 ^a	0.5532 ± 0.0026 ^a
w/o Propedia	54.2M	0.6109 ± 0.0038	<u>0.2713 ± 0.0047</u>	0.1259 ± 0.0105	0.5781 ± 0.0067

^aSignificantly different from HELM-BERT on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 5$). Metrics: area under the receiver operating characteristic curve (ROC-AUC), area under the precision–recall curve (PR-AUC), Matthews correlation coefficient (MCC),^{33,34} and balanced accuracy. SMILES-HELM-BERT is the architecture-matched DeBERTa-style SMILES control defined in Table 3. “w/o Propedia” is HELM-BERT pretrained after excluding Propedia from the pretraining corpus. ^bRandom 5-fold cross-validation. Params indicates peptide encoder parameters. Best results are **bolded**, second-best are underlined. ↑: higher is better.

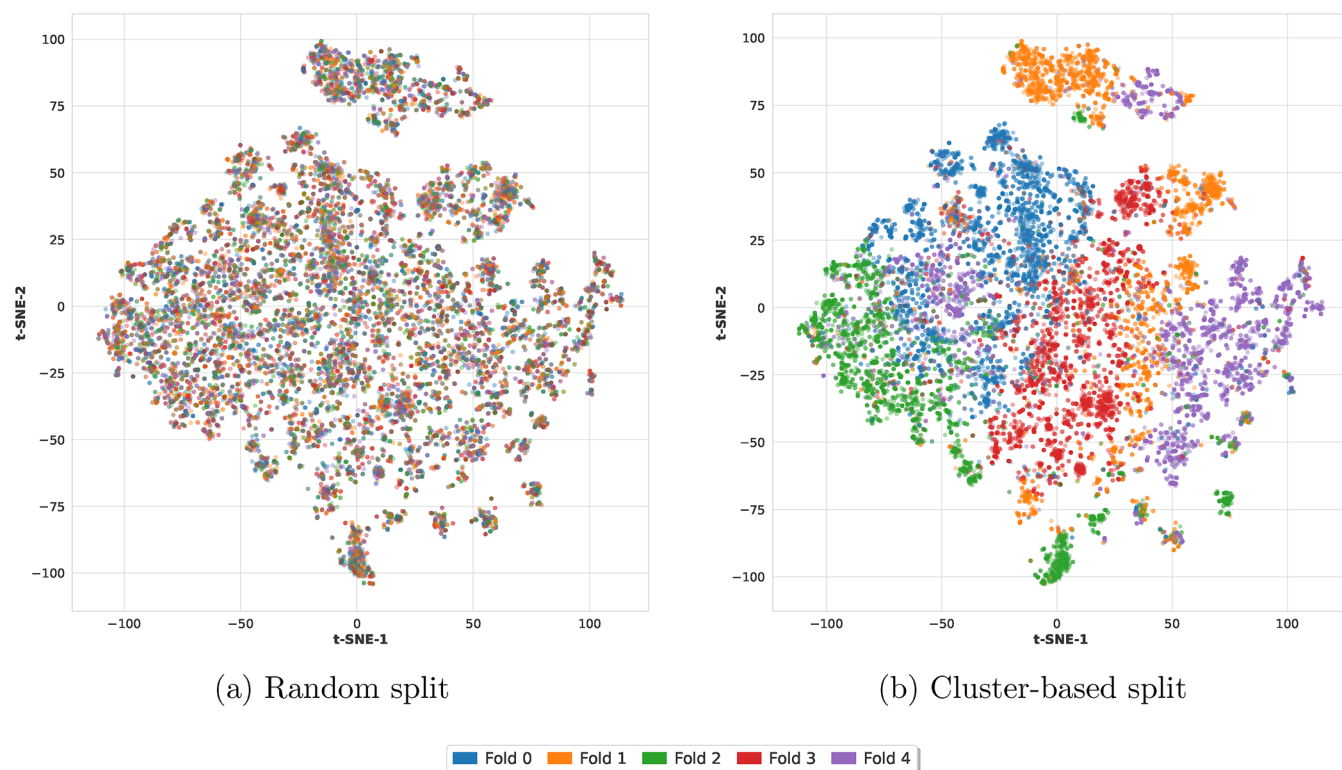


Figure 6. t-SNE projections of PPI data set splits in aCSM complex space. (a) Random split and (b) cluster-based split. Colors indicate fold assignment. The cluster-based split shows distinct spatial separation between folds, reflecting heterogeneous complex-cluster distributions.

HELM Representations Generalize to Peptide–Protein Interaction Prediction

To test generalization beyond permeability and beyond the pretraining objective, we evaluated HELM-BERT embeddings

on peptide–protein interaction (PPI) prediction using Propedia-derived peptide–protein complexes. This task probes whether HELM-BERT peptide representations transfer to interactions with structured protein partners in a standard-

Table 9. Performance Comparison Using Propedia PPI Dataset with aCSM (Cluster-Based) Split^{a,b}

model	params	ROC-AUC ↑	PR-AUC ↑	MCC ↑	Bal. Acc ↑
MLP					
HELM-BERT	54.2M	0.7713 ± 0.0454	0.5063 ± 0.0632	0.3276 ± 0.0654	0.6915 ± 0.0390
ESM-2 (650M)	651M	0.7885 ± 0.0339	<u>0.5263 ± 0.0397</u>	0.3356 ± 0.0545	<u>0.6989 ± 0.0339</u>
ESM-2 (150M)	148M	0.7789 ± 0.0473	0.5118 ± 0.0718	<u>0.3367 ± 0.0775</u>	0.6971 ± 0.0426
ESM-2 (35M)	34M	<u>0.7882 ± 0.0424</u>	0.5282 ± 0.0561	0.3464 ± 0.0669	0.7046 ± 0.0330
PeptideCLM	43.0M	0.7418 ± 0.0264	0.4516 ± 0.0353	0.2798 ± 0.0377	0.6712 ± 0.0227
MoLFormer-XL	44.4M	0.7516 ± 0.0378	0.4643 ± 0.0496	0.2938 ± 0.0588	0.6795 ± 0.0330
Peptide descriptors	—	0.7389 ± 0.0572	0.4627 ± 0.0763	0.2863 ± 0.0762	0.6700 ± 0.0471
SMILES-HELM-BERT	54.2M	0.7454 ± 0.0344	0.4570 ± 0.0501	0.2820 ± 0.0508	0.6724 ± 0.0299
w/o Propedia	54.2M	0.7580 ± 0.0470	0.4762 ± 0.0635	0.3052 ± 0.0654	0.6823 ± 0.0379
Linear					
HELM-BERT	54.2M	0.5549 ± 0.0225	0.2304 ± 0.0158	0.0580 ± 0.0310	0.5336 ± 0.0183
ESM-2 (650M)	651M	0.5644 ± 0.0195	<u>0.2276 ± 0.0071</u>	0.0647 ± 0.0130	0.5373 ± 0.0068
ESM-2 (150M)	148M	<u>0.5587 ± 0.0160</u>	0.2259 ± 0.0107	0.0590 ± 0.0161	0.5354 ± 0.0086
ESM-2 (35M)	34M	0.5517 ± 0.0123	0.2227 ± 0.0099	0.0520 ± 0.0232	0.5317 ± 0.0148
PeptideCLM	43.0M	0.5415 ± 0.0106	0.2234 ± 0.0119	0.0414 ± 0.0139	0.5250 ± 0.0090
MoLFormer-XL	44.4M	0.5484 ± 0.0142	0.2237 ± 0.0087	<u>0.0599 ± 0.0198</u>	<u>0.5366 ± 0.0123</u>
Peptide descriptors	—	0.5310 ± 0.0201	0.2117 ± 0.0093	0.0274 ± 0.0344	0.5170 ± 0.0210
SMILES-HELM-BERT	54.2M	0.5279 ± 0.0099	0.2162 ± 0.0087	0.0255 ± 0.0189	0.5156 ± 0.0118
w/o Propedia	54.2M	0.5504 ± 0.0226	0.2259 ± 0.0126	0.0529 ± 0.0253	0.5312 ± 0.0154

^aSignificantly different from HELM-BERT on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 5$). Metrics: area under the receiver operating characteristic curve (ROC-AUC), area under the precision–recall curve (PR-AUC), Matthews correlation coefficient (MCC), and balanced accuracy. SMILES-HELM-BERT is the architecture-matched DeBERTa-style SMILES control defined in Table 3. “w/o Propedia” is HELM-BERT pretrained after excluding Propedia from the pretraining corpus. ^baCSM cluster-based 5-fold cross-validation: peptide–protein pairs are clustered by aCSM complex signatures and clusters are held out together. Params indicates peptide encoder parameters. Best results are **bolded**, second-best are underlined. ↑: higher is better.

amino-acid peptide setting, and we used two evaluation regimes: a random split and a cluster-based split designed to reduce information leakage through shared peptide–protein complex contact patterns.

Under the random split, HELM-BERT achieved strong performance with an MLP head (ROC-AUC = 0.9419 ± 0.0070 ; Table 8). Compared with SMILES-based peptide encoders (MoLFormer-XL and PeptideCLM), HELM-BERT showed large paired effect sizes under head finetuning ($d = 1.50$ and $d = 2.05$, respectively; Tables 8 and S11), though after panel-wide Benjamini–Hochberg correction across all 32 baseline–metric cells in the Random/MLP panel, neither comparison reached FDR significance (MoLFormer-XL $p_{\text{adj}} = 0.32$; PeptideCLM $p_{\text{adj}} = 0.20$). Under linear probing, which isolates representation quality, HELM-BERT (ROC-AUC = 0.6167 ± 0.0059) significantly outperformed both MoLFormer-XL (0.5949 ± 0.0047) and PeptideCLM (0.5956 ± 0.0045) with paired $d = 2.90$ ($p_{\text{adj}} = 0.025$) and $d = 3.77$ ($p_{\text{adj}} = 0.015$), respectively (Table S12). Fold-wise results confirm consistent performance patterns across all folds (Table S13).

We further evaluated a cluster-based split, where each fold tests distinct complex clusters defined by aCSM signatures (Figure 6). Under head finetuning, HELM-BERT (ROC-AUC = 0.7713 ± 0.0454) outperformed MoLFormer-XL (0.7516 ± 0.0378) and PeptideCLM (0.7418 ± 0.0264) in mean, with paired effect sizes $d = 1.91$ and $d = 1.42$, respectively; no pairwise comparison reached FDR significance under this split (Table S14). Under linear probing, HELM-BERT again showed higher mean performance than MoLFormer-XL (0.5484 ± 0.0142) and PeptideCLM (0.5415 ± 0.0106), with paired effect sizes $d = 0.58$ and $d = 0.68$; again no pairwise comparison reached FDR significance (Table S15). Fold-wise results reflect the heterogeneous test distributions from cluster-

based partitioning (Table S16). Across both splits, peptide descriptors remained below HELM-BERT and the ESM-2 baselines, and they were particularly weak under linear probing, indicating that learned representations capture interaction-relevant features beyond hand-crafted properties.

Because the Propedia PPI data set contains exclusively linear peptides composed of standard amino acids after preprocessing, a setting particularly favorable to protein language models, we also compared HELM-BERT with ESM-2 variants whose evolutionary representations were trained on millions of protein sequences comprising standard residues. Under the random split, ESM-2 variants achieved comparable or slightly higher MLP performance (ROC-AUC = 0.942 – 0.947 vs. 0.942), while under the cluster-based split ESM-2 showed moderately higher MLP scores (ROC-AUC = 0.779 – 0.789 vs. 0.771 ; Tables 8 and 9). Under linear probing, ESM-2 (650M) and HELM-BERT showed comparable random-split ROC-AUC (0.6205 ± 0.0026 vs. 0.6167 ± 0.0059 ; paired $d = -0.87$, $p_{\text{adj}} = 0.31$), with no FDR-significant difference (Table S12). Similarly in the cluster-based split, HELM-BERT and ESM-2 (650M) were not significantly different in ROC-AUC (0.5549 ± 0.0225 vs. 0.5644 ± 0.0195 ; paired $d = -1.02$, $p_{\text{adj}} = 0.82$; Table S15). Despite being pretrained on only $\sim 39,000$ sequences, HELM-BERT produced peptide representations competitive with evolutionary protein embeddings trained on orders of magnitude more data. Moreover, unlike protein-only encoders, HELM-BERT can represent nonstandard residues and chemical modifications directly in HELM, positioning it as a complementary peptide-side representation for interaction modeling. An ablated HELM-BERT variant pretrained without Propedia (“w/o Propedia”) showed no statistically significant difference from the full model on any metric under either split or either head (all $p_{\text{adj}} > 0.13$; Tables S11, S12, S14, and S15),

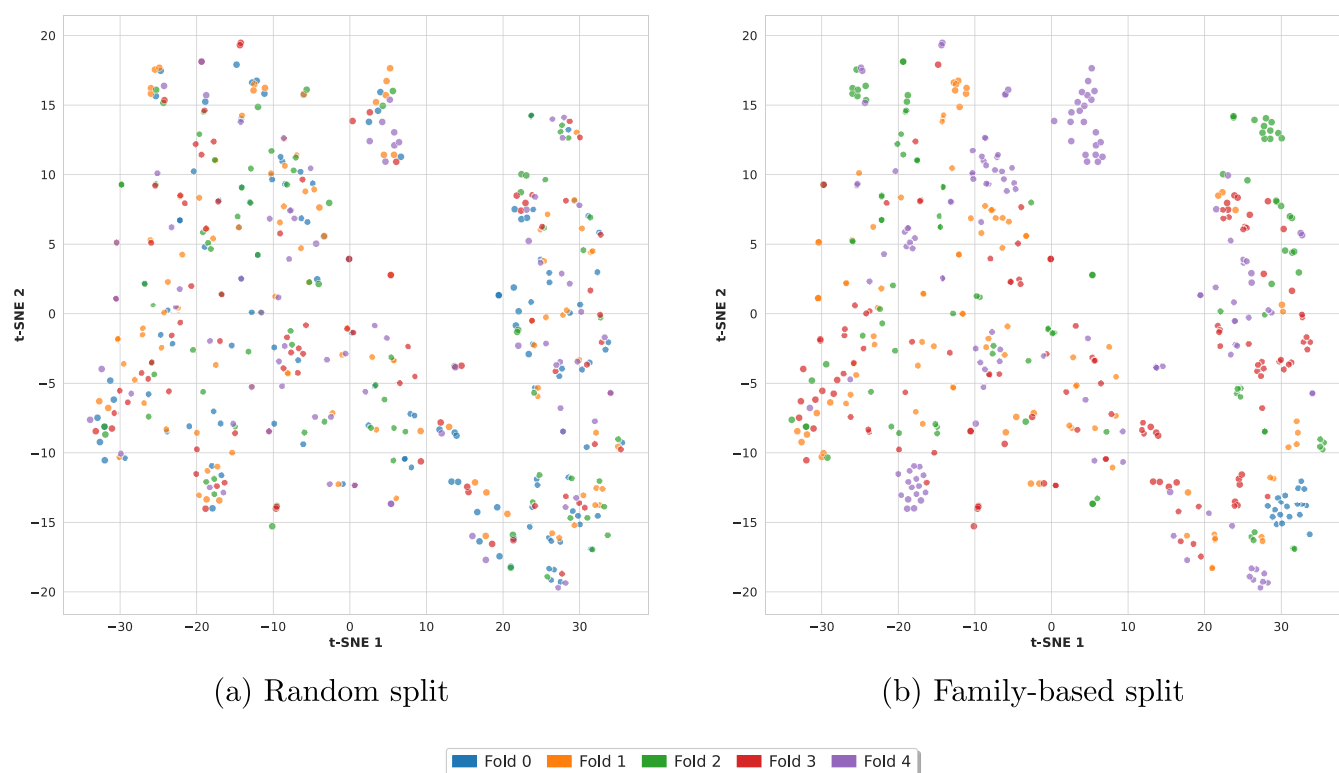


Figure 7. t-SNE projections of ChEMBL PPI data set splits. (a) Random 5-fold split and (b) family based 5-fold split. Colors indicate fold assignment. The family based split yields spatially distinct fold regions driven by protein–target family structure, providing a more stringent generalization test than the random split.

Table 10. Performance on the ChEMBL Peptide–Protein Interaction Benchmark (Random Split)^b

model	ROC-AUC ↑	PR-AUC ↑	MCC ↑	Bal. Acc ↑
MLP				
HELM-BERT	0.9884 ± 0.0019	<u>0.9553 ± 0.0072</u>	<u>0.8811 ± 0.0172</u>	0.9585 ± 0.0058
MoLFormer–XL	0.9867 ± 0.0028	0.9508 ± 0.0076	0.8766 ± 0.0185	0.9554 ± 0.0054
PeptideCLM	0.9849 ± 0.0018	0.9458 ± 0.0047	0.8581 ± 0.0199	0.9493 ± 0.0040
SMILES-HELM-BERT	0.9825 ± 0.0014	0.9400 ± 0.0035	0.8431 ± 0.0170	0.9443 ± 0.0031
w/o ChEMBL	<u>0.9882 ± 0.0016</u>	0.9566 ± 0.0054	0.8919 ± 0.0131	<u>0.9576 ± 0.0059</u>
Linear Probing				
HELM-BERT	0.8680 ± 0.0035	0.5725 ± 0.0039	0.5217 ± 0.0093	0.8021 ± 0.0051
MoLFormer–XL	0.8771 ± 0.0053^a	0.5872 ± 0.0077 ^a	0.5365 ± 0.0086	0.8133 ± 0.0055
PeptideCLM	<u>0.8766 ± 0.0026^a</u>	<u>0.5950 ± 0.0067^a</u>	<u>0.5314 ± 0.0081</u>	<u>0.8107 ± 0.0037</u>
SMILES-HELM-BERT	0.8564 ± 0.0037 ^a	0.5953 ± 0.0082^a	0.4822 ± 0.0067 ^a	0.7796 ± 0.0036 ^a
w/o ChEMBL	0.8663 ± 0.0038 ^a	0.5770 ± 0.0043	0.5167 ± 0.0075	0.7990 ± 0.0036

^aSignificantly different from HELM-BERT on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 5$). Metrics: area under the receiver operating characteristic curve (ROC-AUC), area under the precision–recall curve (PR-AUC), Matthews correlation coefficient (MCC), and balanced accuracy. SMILES-HELM-BERT is the architecture-matched DeBERTa-style SMILES control defined in Table 3. “w/o ChEMBL” is HELM-BERT pretrained after excluding ChEMBL from the pretraining corpus. ^bRandom 5-fold cross-validation. Best results are **bolded**, second-best are underlined. ↑: higher is better.

indicating no evidence of a large corpus-level benchmark-overlap advantage from including Propedia in the pretraining corpus.

Generalization to a Second PPI Benchmark (ChEMBL)

To test whether the PPI results transfer beyond the Propedia data set, we additionally evaluated HELM-BERT on a larger peptide–protein interaction benchmark derived from ChEMBL. We used two complementary splits: a random 5-fold split and a family based split in which pairs sharing the same protein target family, defined by the full hierarchical path of ChEMBL’s `protein_class_desc` field (with unan-

notated targets pooled into a single unclassified group; see Methods), are held out of training (Figure 7). In this benchmark we also report results for SMILES-HELM-BERT (an architecture-matched DeBERTa-style encoder pretrained on canonical SMILES tokens) and for an ablated HELM-BERT variant pretrained without ChEMBL (“w/o ChEMBL”), in order to separate representation capacity from potential benchmark overlap. For each baseline peptide encoder (MoLFormer–XL, PeptideCLM), peptide embeddings were concatenated with ESM-2 (650M) protein embeddings, matching the dual-encoder framework used for HELM-BERT throughout this benchmark.

Table 11. Performance on the ChEMBL Peptide–Protein Interaction Benchmark (Family-Based Split)^{a,b}

model	ROC-AUC ↑	PR-AUC ↑	MCC ↑	Bal. Acc ↑
MLP				
HELM-BERT	0.7808 ± 0.1025	0.5204 ± 0.1231	0.3463 ± 0.1446	0.6752 ± 0.0821
MoLFormer–XL	<u>0.7518 ± 0.0782</u>	0.4419 ± 0.0924	0.3063 ± 0.1220	<u>0.6551 ± 0.0695</u>
PeptideCLM	0.7454 ± 0.0703	0.4419 ± 0.0877	0.2711 ± 0.1289	0.6417 ± 0.0798
SMILES-HELM-BERT	0.7375 ± 0.0953	0.4343 ± 0.1040	0.2668 ± 0.1608	0.6507 ± 0.0911
w/o ChEMBL	0.7408 ± 0.0978	<u>0.4565 ± 0.1146</u>	<u>0.3120 ± 0.1387</u>	0.6534 ± 0.0790
Linear Probing				
HELM-BERT	0.6589 ± 0.1252	0.3477 ± 0.0740	<u>0.2202 ± 0.1377</u>	<u>0.6266 ± 0.0823</u>
MoLFormer–XL	0.6443 ± 0.1069	0.3078 ± 0.0580	0.1815 ± 0.0821	0.6047 ± 0.0513
PeptideCLM	0.6486 ± 0.1171	0.3230 ± 0.0751	0.1880 ± 0.0886	0.6055 ± 0.0543
SMILES-HELM-BERT	<u>0.6504 ± 0.1317</u>	0.3916 ± 0.1014	0.2301 ± 0.1555	0.6297 ± 0.0920
w/o ChEMBL	0.6474 ± 0.1263	<u>0.3680 ± 0.0800</u>	0.2131 ± 0.1361	0.6200 ± 0.0798

^aSignificantly different from HELM-BERT on that metric (corrected resampled *t*-test with Benjamini–Hochberg FDR correction within each panel, $p_{\text{adj}} < 0.05$, $k = 5$). Metrics: area under the receiver operating characteristic curve (ROC-AUC), area under the precision–recall curve (PR-AUC), Matthews correlation coefficient (MCC), and balanced accuracy. SMILES-HELM-BERT is the architecture-matched DeBERTa-style SMILES control defined in Table 3. “w/o ChEMBL” is HELM-BERT pretrained after excluding ChEMBL from the pretraining corpus. ^bFamily based 5-fold cross-validation: pairs whose protein targets share the same ChEMBL protein-family annotation are held out together. Best results are **bolded**, second-best are underlined. ↑: higher is better.

Under the random split (Table 10), all encoders reached uniformly high MLP performance (ROC-AUC $\in [0.983, 0.988]$), and no pairwise comparison between HELM-BERT and any baseline reached FDR significance on ROC-AUC. HELM-BERT nonetheless achieved the highest mean ROC-AUC (0.9884 ± 0.0019) and was indistinguishable from its “w/o ChEMBL” variant (0.9882 ± 0.0016 ; $d = 0.14$, $p_{\text{adj}} = 0.85$), indicating that the strong random-split performance is not attributable to benchmark overlap with the pretraining corpus. Under linear probing on the random split, differences sharpened: HELM-BERT (0.8680 ± 0.0035) significantly outperformed SMILES-HELM-BERT (0.8564 ± 0.0037 ; $d = 8.12$, $p_{\text{adj}} < 10^{-3}$) and its own “w/o ChEMBL” variant (0.8663 ± 0.0038 ; $d = 3.43$, $p_{\text{adj}} = 0.007$), while MoLFormer–XL and PeptideCLM showed small but FDR-significant advantages over HELM-BERT on ROC-AUC ($d = -3.84$ and -8.70 , $p_{\text{adj}} \leq 0.006$); both baselines were paired with ESM-2 in the same dual-encoder framework as HELM-BERT throughout this benchmark. Under ChEMBL linear probing, HELM-BERT is therefore not uniformly best: it outperforms the architecture-matched SMILES-HELM-BERT on ROC-AUC, MCC, and balanced accuracy, is outperformed by SMILES-HELM-BERT on PR-AUC, and is outperformed by MoLFormer–XL and PeptideCLM on ROC-AUC.

Under the family based split, which enforces that no protein family is shared between training and test folds, absolute performance dropped sharply and variability increased across folds (Table 11). HELM-BERT achieved the highest mean ROC-AUC under both the MLP head (0.7808 ± 0.1025) and the linear probe (0.6589 ± 0.1252), but no pairwise comparison reached FDR significance at $k = 5$ folds (all $p_{\text{adj}} > 0.19$; Tables S20 and S21). The large fold-level variance is consistent with the heterogeneous test distributions induced by holding out entire protein families. Overall, the ChEMBL benchmark corroborates the Propedia results: HELM-BERT produces competitive PPI performance on a second, larger data set, with the highest mean ROC-AUC under the family based split but without FDR-significant pairwise separation at $k = 5$ folds. Per-comparison statistics and fold-wise results are reported in Tables S17–S22.

DISCUSSION

This study develops and evaluates a transferable representation for chemically modified and topologically constrained peptides by pretraining an encoder-only transformer directly on HELM notation. The core advance is representational: modified and cyclic peptides occupy a space that is neither well served by atom-level chemical strings nor residue-level protein sequences. By using a notation that explicitly specifies monomer identity and covalent connectivity, HELM-BERT learns embeddings in which chemical modification and HELM-specified topology remain accessible, enabling systematic transfer across peptide tasks.

In permeability prediction on the random split, HELM-BERT achieved the highest mean performance across all three training protocols. It significantly outperformed MoLFormer–XL on the primary R^2 metric across full finetuning, head finetuning, and linear probing, and significantly outperformed PeptideCLM under full and head finetuning; the linear-probe gap to PeptideCLM remained positive but marginal after FDR correction ($d = 0.85$, $p_{\text{adj}} = 0.108$). HELM-BERT also significantly outperformed the architecture-matched SMILES-HELM-BERT control under head finetuning and linear probing, whereas the full-finetuning gap to this matched control narrowed and was not FDR-significant. Because SMILES-HELM-BERT uses the same DeBERTa-style architecture, pretraining corpus, MLM objective, optimizer, and training schedule as HELM-BERT, the matched comparison provides a stricter estimate of the remaining input-representation/tokenization contribution after controlling for architecture and pretraining recipe, and shows that this contribution becomes small under full finetuning. When the entire encoder is free to update, downstream optimization can partially compensate for differences in the input representation; the stronger head finetuning and linear probing gaps suggest that HELM notation/tokenization makes permeability-relevant features more linearly accessible when the encoder cannot fully adapt. Under the Murcko scaffold split, HELM-BERT retained the best mean performance across all metrics and training protocols. The advantage over external SMILES-based encoders was FDR-significant under full finetuning, while the frozen-encoder settings showed positive mean gaps

but weaker statistical separation because of higher fold-level variance. Thus, the scaffold split supports HELM-BERT's robustness under distribution shift, while also showing that the strength of statistical separation depends on finetuning capacity and fold heterogeneity. Notably, even when predictive accuracy converges under full finetuning, HELM-BERT maintains a concrete computational advantage: on the same 39,079-peptide corpus with matched DeBERTa-style encoders and batch size, full finetuning on CycPeptMPDB was approximately $2.2 \times$ faster than SMILES-HELM-BERT (Table S23), underscoring that shorter compressed HELM tokenization provides efficiency gains alongside the representation-level advantages observed in frozen-representation settings.

Architecturally, no single component of HELM-BERT (disentangled attention, EMD, nGiE, span masking, or explicit connectivity tokens) reached FDR significance in isolation on the downstream permeability task; only the cumulative replacement of the DeBERTa-style architecture/objective stack with Vanilla-BERT was FDR-significant, indicating that these components contribute jointly rather than through a single dominant mechanism. However, disentangled attention had a clearer effect on pretraining itself: removing it destabilized MLM training (64% more epochs to convergence, 70% higher terminal validation loss), and the observed compensatory increase in absolute position-embedding norms is consistent with partial compensation by absolute position embeddings when relative-position modeling is removed during MLM pretraining.

Beyond predictive performance, we show that topology information explicitly provided in HELM remains linearly accessible in the embedding space. HELM-BERT embeddings support near-perfect linear separability of structure type and number of connections, and low-dimensional projections reveal clearer structural clustering than SMILES-based peptide encoders. These topology labels are derived from HELM connectivity annotations. Therefore, this analysis is best interpreted as a representation-accessibility check rather than as independent evidence of de novo structural understanding. We note, however, that preserving explicit topology information through to the final embedding is not trivial: SMILES-based encoders receive atom-level molecular connectivity in their inputs but retain these HELM-derived topology labels less accessibly in their embeddings. We therefore view this preservation as a desirable property of HELM-BERT in its own right.

In PPI prediction on the Propedia benchmark, HELM-BERT achieved performance comparable to large protein language model embeddings while requiring orders of magnitude fewer pretraining samples. This result should be interpreted as standard-peptide competitiveness/complementarity, because Propedia contains linear peptides composed of standard amino acids after preprocessing. These findings extend to a second, larger peptide–protein interaction benchmark constructed from ChEMBL, in which nonstandard monomers are widespread at the peptide level but more moderate at the monomer-token level: under the random split, HELM-BERT reached an MLP ROC-AUC of 0.988 ± 0.002 , with no pairwise FDR-significant difference from any baseline or from its “w/o ChEMBL” variant. Under linear probing on the random split, HELM-BERT was significantly better than the architecture-matched SMILES-HELM-BERT on ROC-AUC, MCC, and balanced accuracy, and significantly better

than its “w/o ChEMBL” variant on ROC-AUC; however, HELM-BERT was outperformed by SMILES-HELM-BERT on PR-AUC and by MolFormer-XL and PeptideCLM on ROC-AUC (all models use the same dual-encoder framework with ESM-2 as the frozen protein encoder, so these differences reflect the choice of peptide-side encoder). Under the family based split, HELM-BERT retained the highest mean ROC-AUC across both heads, whereas other metrics showed no FDR-significant differences and in some cases favored SMILES-HELM-BERT. Together, CycPeptMPDB remains the direct modified/cyclic-peptide test, Propedia PPI is a standard-peptide competitiveness benchmark, and ChEMBL PPI provides a nonstandard-monomer-rich PPI benchmark with an explicit benchmark-source overlap control. Across these complementary regimes, HELM-BERT is best viewed as competitive and complementary rather than uniformly superior. Despite being pretrained on only $\sim 39,079$ sequences—orders of magnitude smaller than the corpora used to train ESM-2—HELM-BERT achieves competitive PPI performance, suggesting that topology-explicit pretraining can yield data-efficient and broadly transferable peptide representations. Moreover, because HELM-BERT can directly represent nonstandard residues and chemical modifications, it can serve as a peptide-side representation in interaction modeling settings where protein-only encoders are structurally limited.

Beyond predictive performance, the HELM-vs-SMILES comparison has systematic implications for training cost. Under matched DeBERTa-style encoders, identical corpus, MLM objective, optimizer, schedule, batch size, and hardware, compressed HELM tokenization uses a maximum sequence length of 512 tokens versus 1024 for canonical SMILES, with dynamic padding applied within both ceilings (each batch is padded to the longest sequence in that batch rather than to the global maximum), so that the reported wall-clock values reflect actual per-iteration compute under comparable implementations. Because self-attention scales quadratically with sequence length, this representational difference translates into substantial wall-clock gains during the training stages that update the peptide encoder: MLM pretraining on HELM was approximately $14 \times$ faster per epoch than on SMILES (1.42 vs. 20.1 min/epoch), and downstream full finetuning on CycPeptMPDB permeability was approximately $2.2 \times$ faster under both the random (324.6 vs. 738.0 min) and Murcko scaffold (311.6 vs. 688.1 min) splits (Table S23). This timing analysis is intended as a compute-efficiency comparison rather than as additional evidence of universal HELM superiority. Across the downstream evaluations reported here (permeability prediction under full, head, and linear probing in Tables 3 and 4; peptide–protein interaction prediction on Propedia and ChEMBL in Tables 8–11), HELM-BERT matches or exceeds the architecture-matched SMILES-HELM-BERT control on most metrics, with SMILES-HELM-BERT retaining a small PR-AUC advantage under ChEMBL linear probing; the compute gain is therefore obtained while maintaining or improving predictive performance on the majority of evaluations. A complementary memory-side consequence is that HELM-BERT trained at effective batch 64 in a single forward/backward pass on a single GPU, whereas SMILES-HELM-BERT required gradient accumulation (32×2) to reach the same effective batch; this reduces the hardware requirement for matched-setup training. Linear probing and PPI finetuning are excluded from the timing comparison

because the peptide encoder is frozen in both settings, so their wall-clock is dominated by scikit-learn regularization-strength grid search and by cached-embedding head training rather than by peptide-encoder compute. These observations support compressed HELM tokenization as a compute-efficient input representation for pretraining on chemically modified peptides.

This work has several limitations. First, HELM-based approaches rely on the availability and correctness of HELM annotations; although conversion from structural sources is possible, coverage of complex modifications is incomplete in many databases. Second, while the current pretraining corpus enables strong transfer, scaling to larger and more diverse HELM-annotated corpora may further improve generalization. Third, the permeability benchmark would benefit from topology-held-out splits beyond Murcko scaffolds (e.g., withholding specific cyclization motifs, ring topologies, linker chemistries, or classes of noncanonical monomers). Finally, the current architecture focuses on encoder-only representations; future hybrid models could combine HELM-level polymer topology with atom-level monomer descriptors, graph-derived monomer features, or structure-informed objectives, and PPI-specific extensions may benefit from larger interaction-relevant peptide corpora and protein-interface-aware architectures.

More broadly, this work suggests a general recipe for modeling complex biomolecular modalities that lie between “chemistry” and “sequence”: choose a representation where task-relevant structural constraints (here, connectivity and topology) are explicit, and pretrain an encoder whose inductive biases preserve and use those constraints. For peptide therapeutics, topology-aware representations can reduce reliance on task-specific feature engineering and provide a unified starting point for prediction across modified, cyclic, and cross-linked peptides.

■ ASSOCIATED CONTENT

Data Availability Statement

The training and evaluation code for HELM-BERT, together with the curated pretraining corpus and processed downstream data sets, is available from the project repository at <https://github.com/clinfo/HELM-BERT>. CycPeptMPDB, Propedia, and ChEMBL are available from their original distributions, as cited in the References. Pretrained model weights are released through the Hugging Face model hub at <https://huggingface.co/Flansma/helm-bert>.

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.6c00451>.

PPI prediction framework; statistical comparison of models; fold-wise performance results; HELM-BERT ablation studies; component activation analysis; embedding quality analysis; low-dimensional embedding analysis; PPI prediction statistical comparisons; embedding visualizations colored by structure type, source data set, and physicochemical properties (PDF)

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Author Contributions

T.K. and S.M. conceptualized the study. S.L. developed the methodology, implemented the software, performed the experiments, and analyzed the data. S.L. and T.K. were responsible for data curation. S.L. wrote the original draft of the manuscript. T.K., I.M., S.M., and Y.O. supervised the study and reviewed and edited the manuscript. S.M. and Y.O. supervised the study and acquired funding. All the authors have read and agreed to the published version of the manuscript.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research (in part) used computational resources of the supercomputer Miyabi provided by the Information Technology Center, The University of Tokyo. This work was supported by the Japan Agency for Medical Research and Development (AMED) under Grant Number JP25nk0101112.

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